10

Stability of static MHD equilibria

10.1 Introduction

Solutions to Eq. (9.50), the Grad–Shafranov equation, (or to some more complicated counterpart in the case of non-axisymmetric geometry) provide a static MHD equilibrium. The question now arises whether the equilibrium is stable. This issue was forced upon early magnetic fusion researchers who found that plasma that was expected to be well confined in a static MHD equilibrium configuration would instead became violently unstable and crash destructively into the wall in a few microseconds.

The difference between stable and unstable equilibria is shown schematically in Fig. 10.1. Here a ball, representing the plasma, is located at either the bottom of a valley or the top of a hill. If the ball is at the bottom of a valley, i.e., a minimum in the potential energy, then a slight lateral displacement results in a restoring force, which pushes the ball back. The ball then overshoots and oscillates about the minimum with a constant amplitude because energy is conserved. On the other hand, if the ball is initially located at the top of a hill, then a slight lateral displacement results in a force that pushes the ball further to the side so that there is an increase in the velocity. The perturbed force is not restoring, but rather the opposite. The velocity is always in the direction of the original displacement; i.e., there is no oscillation in velocity.

The equation of motion for this system is

$$m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = \pm\kappa x, \tag{10.1}$$

where κ is assumed positive; the plus sign is chosen for the ball on hill case, and the minus sign is chosen for the ball in valley case. This equation has a solution $x \sim \exp(-\mathrm{i}\omega t)$, where $\omega = \pm\sqrt{\kappa/m}$ for the valley case and $\omega = \pm\mathrm{i}\sqrt{\kappa/m}$ for the hill case. The hill $\omega = +\mathrm{i}\sqrt{\kappa/m}$ solution is unstable and corresponds to the ball accelerating down the hill when perturbed from its initial equilibrium position.

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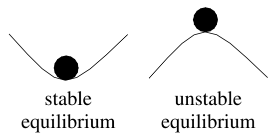

Fig. 10.1 Stable and unstable equilibria.

If this configuration is extended to two dimensions, then stability would require an absolute minimum in both directions. However, a saddle point potential would suffice for instability, since the ball could always roll down from the saddle point. Thus, a multi-dimensional system can only be stable if the equilibrium potential energy corresponds to an absolute minimum with respect to all possible displacements.

The problem of determining MHD stability is analogous to having a ball on a multi-dimensional hill. If the potential energy of the system increases for any allowed perturbation of the system, then the system is stable. However, if there exists even a single allowed perturbation that decreases the system's potential energy, then the system is unstable.

10.2 The Rayleigh–Taylor instability of hydrodynamics

An important subset of MHD instabilities is formally similar to the Rayleigh–Taylor instability of hydrodynamics; it is therefore useful to put aside MHD for the moment and examine this classical problem. As any toddler learns from stacking building blocks, it is possible to construct an equilibrium whereby a heavy object is supported by a light object, but such an equilibrium is unstable. The corresponding hydrodynamic situation has a heavy fluid supported by a light fluid as shown in Fig. 10.2(a); this situation is unstable with respect to the rippling shown in Fig. 10.2(b). The ripples are unstable because they effectively interchange volume elements of heavy fluid with equivalent volume elements of

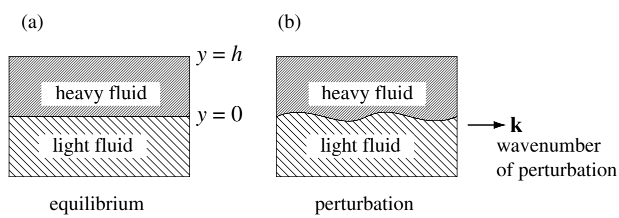

Fig. 10.2 (a) Top-heavy fluid equilibrium, (b) rippling instability.

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light fluid. Each volume element of interchanged heavy fluid originally had its center of mass a distance Δ above the interface while each volume element of interchanged light fluid originally had its center of mass a distance Δ below the interface. Since the potential energy of a mass m at height h in a gravitational field g is mgh, the respective changes in potential energy of the heavy and light fluids are

$$\delta W_h = -2\rho_h V\Delta g, \quad \delta W_l = +2\rho_l V\Delta g, \tag{10.2}$$

where V is the volume of the interchanged fluid elements and ρ_h and ρ_l are the mass densities of the heavy and light fluids. The net change in the total system potential energy is

$$\delta W = -2\left(\rho_h - \rho_l\right) V\Delta g, \tag{10.3}$$

which is negative so the system *lowers* its potential energy by forming ripples. This is analogous to the ball falling off the top of the hill.

A well-known example of this instability is the situation of an inverted glass of water. The heavy fluid in this case is the water and the light fluid is the air. The system is stable when a piece of cardboard is located at the interface between the water and the air, but when the cardboard is removed the system becomes unstable and the water falls out. The function of the cardboard is to prevent ripple interchange from occurring. The system is in stable equilibrium when ripples are prevented because atmospheric pressure is adequate to support the inverted water. From a mathematical point of view, the cardboard places a constraint on the system by imposing a boundary condition that prevents ripple formation.

When the cardboard is removed so that there is no longer a constraint against ripple formation, the ripples grow from noise to large amplitude, and the water falls out. This is an example of an unstable equilibrium.

The geometry shown in Fig. 10.2 is now used to analyze the stability of a heavy fluid such as water supported by a light fluid such as air in the situation where no constraint exists at the interface. Here y corresponds to the vertical direction so that gravity is in the negative y direction. To simplify the analysis, it is assumed that $\rho_l \ll \rho_h$, in which case the mass of the light fluid can be ignored. The water and air are assumed to be incompressible, i.e., $\rho = const.$, so the continuity equation

$$\frac{\partial \rho}{\partial t} + \mathbf{v}\cdot\nabla\rho + \rho\nabla\cdot\mathbf{v} = 0 \tag{10.4}$$

reduces to

$$\nabla\cdot\mathbf{v} = 0. \tag{10.5}$$

The linearized continuity equation in the water therefore reduces to

$$\frac{\partial \rho_1}{\partial t} + \mathbf{v}_1 \cdot \nabla \rho_0 = 0 \tag{10.6}$$

and the linearized equation of motion in the water is

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla P_1 - \rho_1 g \hat{y}. \tag{10.7}$$

The location $y = 0$ is defined to be at the unperturbed air–water interface and the top of the glass is at $y = h$. The water fills the glass to the top and thus is constrained from moving at the top, giving the top boundary condition

$$v_y = 0 \text{ at } y = h. \tag{10.8}$$

The perturbation is assumed to have the form

$$\mathbf{v}_1 = \mathbf{v}_1(y) \mathrm{e}^{\gamma t + \mathrm{i}\mathbf{k}\cdot\mathbf{x}}, \tag{10.9}$$

where $\mathbf{k}$ lies in the $x - z$ plane and positive γ implies instability. The incompressibility condition, Eq. (10.5), can thus be written as

$$\frac{\partial v_{1y}}{\partial y} + \mathrm{i}\mathbf{k} \cdot \mathbf{v}_{1\perp} = 0, \tag{10.10}$$

where $\perp$ means perpendicular to the y direction. The y and $\perp$ components of Eq. (10.7) become respectively

$$\gamma \rho_0 v_{1y} = -\frac{\partial P_1}{\partial y} - \rho_1 g \tag{10.11}$$

$$\gamma \rho_0 \mathbf{v}_{1\perp} = -\mathrm{i}\mathbf{k} P_1. \tag{10.12}$$

The system is solved by dotting Eq. (10.12) with $\mathrm{i}\mathbf{k}$ and then using Eq. (10.10) to eliminate $\mathbf{k} \cdot \mathbf{v}_{1\perp}$ and so obtain

$$-\gamma \rho_0 \frac{\partial v_{1y}}{\partial y} = k^2 P_1. \tag{10.13}$$

The perturbed density, as given by Eq. (10.6), is

$$\gamma \rho_1 = -v_{1y} \frac{\partial \rho_0}{\partial y}. \tag{10.14}$$

Next, ρ_1 and P_1 are substituted for in Eq. (10.11) to obtain the eigenvalue equation

$$\frac{\partial}{\partial y}\left[\gamma^2 \rho_0 \frac{\partial v_{1y}}{\partial y}\right] = \left[\gamma^2 \rho_0 - g \frac{\partial \rho_0}{\partial y}\right] k^2 v_{1y}. \tag{10.15}$$

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This equation is solved by considering the interior and the interface separately:

1. Interior: here $\partial\rho_0/\partial y = 0$ and $\rho_0 = const.$, in which case Eq. (10.15) becomes
$$\frac{\partial^2 v_{1y}}{\partial y^2} = k^2 v_{1y}. \tag{10.16}$$
The solution satisfying the boundary condition given by Eq. (10.8) is
$$v_{1y} = A\sinh(k(y-h)). \tag{10.17}$$
2. Interface: to find the properties of this region, Eq. (10.15) is integrated across the interface from $y = 0_-$ to $y = 0_+$ to obtain
$$\left[\gamma^2\rho_0\frac{\partial v_{1y}}{\partial y}\right]_{0_-}^{0_+} = -\left[g\rho_0 k^2 v_{1y}\right]_{0_-}^{0+} \tag{10.18}$$
or
$$\gamma^2\frac{\partial v_{1y}}{\partial y} = -gk^2 v_{1y}, \tag{10.19}$$
where all quantities refer to the upper (water) side of the interface, since by assumption $\rho_0(y = 0_-) \simeq 0$.

Substitution of Eq. (10.17) into Eq. (10.19) gives the dispersion relation
$$\gamma^2 = kg\tanh(k_\perp h), \tag{10.20}$$
which shows that the configuration is always unstable since $\gamma^2 > 0$. Equation (10.20) furthermore shows that short wavelengths are most unstable, but a more detailed analysis taking into account surface tension (which is stronger for shorter wavelengths) would show γ^2 has a maximum at some wavelength. Above this most unstable wavelength, surface tension would decrease the growth rate.

10.3 MHD Rayleigh–Taylor instability

We define a "magnetofluid" as a fluid that satisfies the MHD equations. A given plasma may or may not behave as a magnetofluid, depending on the validity of the MHD approximation for the circumstances of the given plasma. The magnetofluid concept provides a legalism that allows consideration of the implications of MHD without necessarily accepting that these implications are relevant to a specific actual plasma. In effect, the magnetofluid concept can be considered as a tentative model of plasma.

Let us now replace the water in the Rayleigh–Taylor instability by magnetofluid. We further suppose that instead of atmospheric pressure supporting the magnetofluid, a vertical magnetic field gradient balances the downwards gravitational force, i.e., at each y the upward force of $-\nabla B^2/2\mu_0$ supports the

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downward force of the weight of the plasma above. Although gravity is normally unimportant in actual plasmas, the gravitational model is nevertheless quite useful for characterizing situations of practical interest, because gravity can be considered as a proxy for the actual forces that typically have a more complex structure. An important example is centrifugal force associated with thermal particle motion along curved magnetic field lines acting like a gravitational force in the direction of the radius of curvature of these field lines. The curved field with associated centrifugal force due to parallel thermal motion is replaced by a Cartesian geometry model having straight field lines and, perpendicular to the field lines, a gravitational force is invoked to represent the effect of the centrifugal force. The y direction corresponds to the direction of the radius of curvature.

In order for $-\nabla B^2$ to point upwards in the y direction, the magnetic field must depend on y such that its magnitude decreases with increasing y. Furthermore, it is required that $B_y = 0$ so ∇B^2 is perpendicular to the magnetic field and the field can be considered as locally straight (field line curvature has already been taken into account by introducing the fictitious gravity). Thus, the equilibrium magnetic field is assumed to be of the general form

$$\mathbf{B}_0 = B_{x0}(y)\hat{x} + B_{z0}(y)\hat{z}. \tag{10.21}$$

The unit vector associated with the equilibrium field is

$$\hat{B}_0 = \frac{B_{x0}(y)\hat{x} + B_{z0}(y)\hat{z}}{\sqrt{B_{x0}(y)^2 + B_{z0}(y)^2}}. \tag{10.22}$$

For the special case where $B_{x0}(y)$ and $B_{z0}(y)$ are proportional to each other, the field line direction is independent of y, but in the more general case where this is not so, $\hat{B}_0$ depends on y and thus rotates as a function of y. In this latter case, the magnetic field lines are said to be sheared, since adjacent y-layers of field lines are not parallel to each other.

The magnetofluid is assumed incompressible and, using Eq. (9.10), the linearized equation of motion becomes

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla \bar{P}_1 + \frac{\mathbf{B}_0 \cdot \nabla \mathbf{B}_1 + \mathbf{B}_1 \cdot \nabla \mathbf{B}_0}{\mu_0} - \rho_1 g \hat{y}, \tag{10.23}$$

where

$$\bar{P}_1 = P_1 + \frac{\mathbf{B}_0 \cdot \mathbf{B}_1}{\mu_0}$$

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is the perturbation of the combined hydrodynamic and magnetic pressure, i.e., the perturbation of $P+B^2/2\mu_0$. It is again assumed that all quantities vary in the manner of Eq. (10.9) so Eq. (10.23) has the respective y and $\perp$ components

$$\gamma\rho_0 v_{1y} = -\frac{\partial \bar{P}_1}{\partial y} + \frac{\mathrm{i}\left(\mathbf{k}\cdot\mathbf{B}_0\right)B_{1y}}{\mu_0} - \rho_1 g \tag{10.24}$$

$$\gamma\rho_0 \mathbf{v}_{1\perp} = -\mathrm{i}\mathbf{k}\bar{P}_1 + \frac{1}{\mu_0}\left[\mathrm{i}\left(\mathbf{k}\cdot\mathbf{B}_0\right)\mathbf{B}_{1\perp} + B_{1y}\frac{\partial \mathbf{B}_0}{\partial y}\right]. \tag{10.25}$$

In analogy to the glass of water problem, Eq. (10.25) is dotted with $\mathrm{i}\mathbf{k}$ and Eq. (10.10) is invoked to obtain

$$-\gamma\rho_0\frac{\partial v_{1y}}{\partial y} = k^2\bar{P}_1 + \frac{1}{\mu_0}\left[-\left(\mathbf{k}\cdot\mathbf{B}_0\right)\mathbf{k}\cdot\mathbf{B}_{1\perp} + \mathrm{i}B_{1y}\frac{\partial\left(\mathbf{k}\cdot\mathbf{B}_0\right)}{\partial y}\right]. \tag{10.26}$$

Because $\nabla\cdot\mathbf{B}_1 = 0$, the perturbed perpendicular field is

$$\mathrm{i}\mathbf{k}\cdot\mathbf{B}_{1\perp} = -\frac{\partial B_{1y}}{\partial y} \tag{10.27}$$

and so Eq. (10.26) can be recast as

$$k^2\bar{P}_1 = -\gamma\rho_0\frac{\partial v_{1y}}{\partial y} - \frac{1}{\mu_0}\left[-\mathrm{i}\left(\mathbf{k}\cdot\mathbf{B}_0\right)\frac{\partial B_{1y}}{\partial y} + \mathrm{i}B_{1y}\frac{\partial\left(\mathbf{k}\cdot\mathbf{B}_0\right)}{\partial y}\right]. \tag{10.28}$$

Following a procedure analogous to that used in the inverted glass of water problem, $\bar{P}_1$ is eliminated in Eq. (10.24) by substitution of Eq. (10.28) to obtain

$$\begin{aligned}\gamma\rho_0 v_{1y} = &-\frac{1}{k^2}\frac{\partial}{\partial y}\left\{-\gamma\rho_0\frac{\partial v_{1y}}{\partial y} - \frac{1}{\mu_0}\left[-\left(\mathrm{i}\mathbf{k}\cdot\mathbf{B}_0\right)\frac{\partial B_{1y}}{\partial y} + B_{1y}\frac{\partial\left(\mathrm{i}\mathbf{k}\cdot\mathbf{B}_0\right)}{\partial y}\right]\right\}\\ &+\frac{\mathrm{i}\left(\mathbf{k}\cdot\mathbf{B}_0\right)B_{1y}}{\mu_0} - \rho_1 g.\end{aligned} \tag{10.29}$$

To proceed further, it is necessary to know B_{1y}. The complete vector $\mathbf{B}_1$ is found by first linearizing the MHD Ohm's law to obtain

$$\mathbf{E}_1 + \mathbf{v}_1\times\mathbf{B}_0 = 0, \tag{10.30}$$

then taking the curl, and finally using Faraday's law to obtain

$$\gamma\mathbf{B}_1 = \nabla\times\left[\mathbf{v}_1\times\mathbf{B}_0\right]. \tag{10.31}$$

The y component is found by dotting with $\hat{y}$ and then using the vector identity $\nabla\cdot(\mathbf{F}\times\mathbf{G}) = \mathbf{G}\cdot\nabla\times\mathbf{F} - \mathbf{F}\cdot\nabla\times\mathbf{G}$ to obtain

$$\gamma B_{1y} = \hat{y}\cdot\nabla\times\left[\mathbf{v}_1\times\mathbf{B}_0\right] = \nabla\cdot\left[\left(\mathbf{v}_1\times\mathbf{B}_0\right)\times\hat{y}\right] = \nabla\cdot\left[v_{1y}\mathbf{B}_0\right] = \mathrm{i}\mathbf{k}\cdot\mathbf{B}_0 v_{1y}. \tag{10.32}$$

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Substituting into Eq. (10.29) using Eq. (10.32) and Eq. (10.14) and rearranging the order gives

$$\frac{\partial}{\partial y}\left\{\left[\gamma^2\rho_0+\frac{1}{\mu_0}\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2\right]\frac{\partial v_{1y}}{\partial y}\right\}=k^2\left\{\gamma^2\rho_0-g\frac{\partial\rho_0}{\partial y}+\frac{\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2}{\mu_0}\right\}v_{1y},\quad(10.33)$$

which is identical to the inverted glass of water problem if $\mathbf{k}\cdot\mathbf{B}_0=0$.

Rather than have an abrupt interface between a heavy and light fluid as in the glass of water problem, it is assumed that the magnetofluid fills the container between $y=0$ and $y=h$ and that there is a density gradient in the y direction. This situation is more appropriate for a plasma, which would typically have a continuous density gradient. It is assumed that rigid boundaries exist at both $y=0$ and $y=h$ so that $v_{1y}=0$ at both $y=0$ and $y=h$. These boundary conditions mean that rippling is not allowed at $y=0,h$ but could occur in the interior, $0<y<h$. Equation (10.33) is now impossible to solve analytically because all the coefficients are functions of y. However, an approximate understanding for the behavior predicted by this equation can be found by multiplying the equation by v_{1y} and integrating from $y=0$ to $y=h$ to obtain

$$\left[\left\{\gamma^2\rho_0+\frac{1}{\mu_0}\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2\right\}v_{1y}\frac{\partial v_{1y}}{\partial y}\right]_0^h-\int_0^h\left[\gamma^2\rho_0+\frac{1}{\mu_0}\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2\right]\left(\frac{\partial v_{1y}}{\partial y}\right)^2\mathrm{d}y$$
$$=k^2\int_0^h\left\{\gamma^2\rho_0-g\frac{\partial\rho_0}{\partial y}+\frac{\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2}{\mu_0}\right\}v_{1y}^2\mathrm{d}y.\quad(10.34)$$

The integrated term vanishes because of the boundary conditions (which could also have been $\partial v_{1y}/\partial y=0$). Solving for γ^2 gives

$$\gamma^2=\frac{\int_0^h\mathrm{d}y\left[k^2g\frac{\partial\rho_0}{\partial y}v_{1y}^2-\frac{\left(\mathbf{k}\cdot\mathbf{B}_0\right)^2}{\mu_0}\left(k^2v_{1y}^2+\left(\frac{\partial v_{1y}}{\partial y}\right)^2\right)\right]}{\int_0^h\mathrm{d}y\;\rho_0\left[k^2v_{1y}^2+\left(\frac{\partial v_{1y}}{\partial y}\right)^2\right]}.\quad(10.35)$$

If $\mathbf{k}\cdot\mathbf{B}_0=0$ and the density gradient is positive everywhere then $\gamma^2>0$ so there is instability. If the density gradient is negative everywhere except at one stratum with thickness Δy, then the system will be unstable with respect to an interchange at that one "top-heavy" stratum. The velocity will be concentrated at this unstable stratum and so the integrands will vanish everywhere except at the unstable stratum giving a growth rate $\gamma^2\sim g\Delta y\;\rho_0^{-1}\partial\rho_0/\partial y$, where $\partial\rho_0/\partial y$ is the value in the unstable region. This MHD version of the Rayleigh–Taylor instability is called the Kruskal–Schwarzschild instability (Kruskal and Schwarzschild 1954).

Finite $\mathbf{k}\cdot\mathbf{B}_0$ opposes the effect of the destabilizing positive density gradient, reducing the growth rate to $\gamma^2 \sim g\Delta y\ \rho_0^{-1}\,\partial\rho_0/\partial y - (\mathbf{k}\cdot\mathbf{B}_0)^2/\mu_0$. A sufficiently strong field will stabilize the system. However, thermally excited noise excites modes with all possible values of $\mathbf{k}$ and those modes aligned such that $\mathbf{k}\cdot\mathbf{B}_0 = 0$ will not be stabilized. A shear in the magnetic field has the effect of making it possible to have $\mathbf{k}\cdot\mathbf{B}_0(y) = 0$ at only a single value of y. The lack of any spatial dependence of $\mathbf{k}$ results from the translational invariance of the plasma with respect to both x and z implying that $\mathbf{k} = k_x\hat{x} + k_z\hat{z}$ is independent of position. Thus, magnetic shear constrains the instability to a narrow y stratum.

The stabilizing effect of finite $\mathbf{k}\cdot\mathbf{B}_0$ can be understood by considering Eq. (10.32) which shows that the amount of B_{1y} associated with a given v_{1y} is proportional to $\mathbf{k}\cdot\mathbf{B}_0$. Since the original equilibrium field had no y component, introducing a finite B_{1y} corresponds to "plucking" the equilibrium field. The plucking varies sinusoidally along the equilibrium field and stretches the equilibrium field like a plucked violin string. The plucked field has a restoring force, which pulls the field back to its equilibrium position. If the energy associated with plucking the magnetic field exceeds the energy liberated by the interchange of heavy upper magnetofluid with light lower magnetofluid, the mode is stable.

As discussed earlier, the gravitational force in the magnetofluid model represents the centrifugal force resulting from guiding center motion along curved field lines. This leads to the concept of "good" and "bad" curvature illustrated in Fig. 10.3. A plasma has bad curvature if the field lines at the plasma–vacuum boundary have a convex shape as seen by an observer outside the plasma, since in this case the centrifugal force is outwards from the plasma. Bad curvature gives a centrifugal force that can drive interchange instabilities whereas good curvature corresponds to a concave shape so that the centrifugal force is always inwards. Mirror magnetic fields as sketched in Fig. 10.4 have good curvature in the vicinity of the mirrors and bad curvature in the vicinity of the mirror minimum so that a detailed analysis of interchange instabilities requires averaging the "goodness/badness" along the portion of the flux tube experienced by the particle. Cusp magnetic fields (cf. Fig. 10.4) have good curvature everywhere,

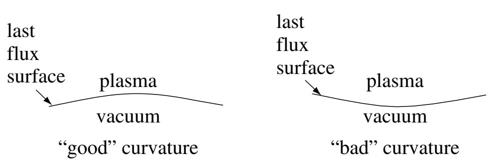

Fig. 10.3 Good and bad curvature of the plasma–vacuum interface.

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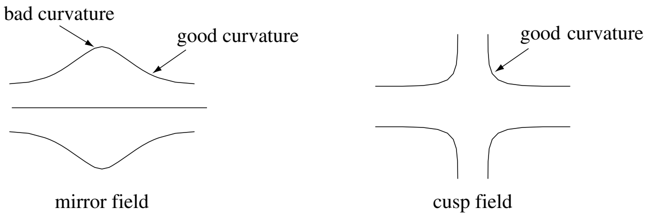

Fig. 10.4 Mirror fields have bad curvature at the mirror minimum, good curvature near the maximum; cusp fields have good curvature everywhere.

but have singular behavior at the cusps. Plasmas with internal currents such as tokamaks have significant shear everywhere, since $\partial B_x/\partial y \sim J_z$ and $\partial B_z/\partial y \sim J_x$; this shear inhibits interchange instabilities. However, as will be shown later these currents can be the source for another class of instability called a current-driven instability.

10.4 The MHD energy principle

Suppose a system initially in equilibrium is subject to a small perturbation instigated by random thermal noise. The perturbation will affect all the various dependent variables in a way that must be consistent with all the relevant equations and boundary conditions. Thus, the perturbation may be considered as a low-level excitation of some allowed normal mode of the system. The mode will be unstable if it reduces the potential energy of the system and the growth rate of the instability will be proportional to the amount by which the potential energy is reduced. In order to investigate whether a given system is stable, it suffices to show that no modes exist that reduce the system's potential energy. Demonstrating MHD stability this way was first done by Bernstein *et al.* (1958) and the method is called the MHD energy principle.

Each mode has its own specific pattern for displacing the magnetofluid volume elements from their equilibrium positions. The pattern of displacements can be represented by a vector function of position $\boldsymbol{\xi}(\mathbf{x})$, which prescribes how a fluid volume element originally at location $\mathbf{x}$ is displaced. Thus, the mode involves moving a fluid element initially at $\mathbf{x}$ to the position $\mathbf{x}+\boldsymbol{\xi}(\mathbf{x})$. The displacement of a magnetofluid element initially at the surface is shown in Fig. 10.5, a sketch of a two-dimensional cut of a three-dimensional magnetofluid (plasma) surrounded by a vacuum region in turn bounded by a conducting wall. The wall could be brought right up to the magnetofluid to eliminate the vacuum region or, alternatively, the wall could be placed at infinity to represent a system having no wall and surrounded by vacuum.

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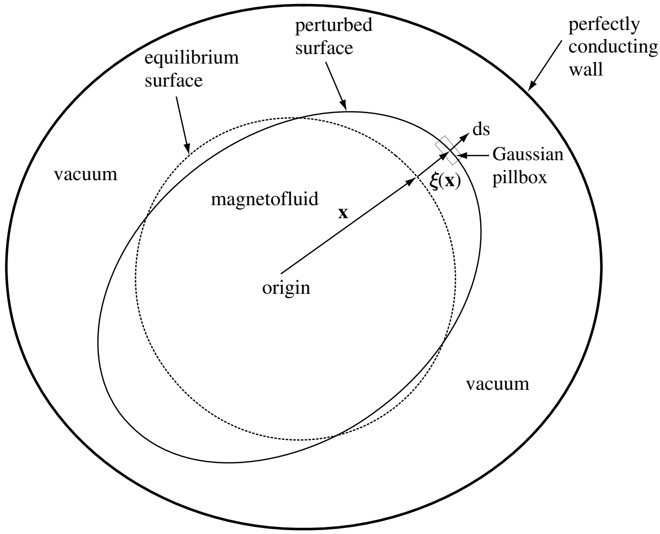

Fig. 10.5 Two-dimensional cut of three-dimensional plasma equilibrium and perturbation; Gaussian pillbox is used to relate quantities in vacuum to quantities in plasma. Each volume element at a position $\mathbf{x}$ is displaced by an amount $\boldsymbol{\xi}(\mathbf{x})$. The displacement of a volume element at the surface is illustrated.

10.4.1 Energy equation for a magnetofluid

The energy content of a magnetofluid can be obtained from the ideal MHD equations if it is assumed that all motions are sufficiently fast to be adiabatic but slow enough for collisions to keep the pressure isotropic. The ideal MHD equations are

$$\rho\left(\frac{\partial \mathbf{U}}{\partial t}+\mathbf{U}\cdot\nabla\mathbf{U}\right)=\mathbf{J}\times\mathbf{B}-\nabla P \tag{10.36}$$

$$\mathbf{E}+\mathbf{U}\times\mathbf{B}=0 \tag{10.37}$$

$$\nabla\times\mathbf{E}=-\frac{\partial \mathbf{B}}{\partial t} \tag{10.38}$$

$$\nabla\times\mathbf{B}=\mu_o\mathbf{J} \tag{10.39}$$

$$\frac{\partial \rho}{\partial t}+\nabla\cdot(\rho\mathbf{U})=0 \tag{10.40}$$

$$P\sim\rho^{\gamma}. \tag{10.41}$$

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Ohm's law, Eq. (10.37), and Faraday's law, Eq. (10.38), are combined to give the *induction equation*

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{U} \times \mathbf{B}), \tag{10.42}$$

which prescribes the magnetic field evolution and, as discussed in the context of Eq. (2.83), shows that the magnetic flux is frozen into the magnetofluid. Equation (10.41), the adiabatic relation, implies the following relationships for spatial and temporal derivatives of density and pressure

$$\frac{\nabla P}{P} = \gamma \frac{\nabla \rho}{\rho}, \quad \frac{1}{P}\frac{\partial P}{\partial t} = \frac{\gamma}{\rho}\frac{\partial \rho}{\partial t}. \tag{10.43}$$

Combining these relationships with the continuity equation, Eq. (10.40), gives the pressure evolution equation

$$\frac{\partial P}{\partial t} + \mathbf{U} \cdot \nabla P + \gamma P \nabla \cdot \mathbf{U} = 0, \tag{10.44}$$

which can also be written as

$$\frac{\partial P}{\partial t} + \nabla \cdot (P\mathbf{U}) + (\gamma - 1) P \nabla \cdot \mathbf{U} = 0. \tag{10.45}$$

An expression for the overall energy can be derived by first writing the equation of motion, Eq. (10.36), as

$$\rho \frac{\partial \mathbf{U}}{\partial t} + \rho \nabla \left(\frac{U^2}{2} \right) - \rho \mathbf{U} \times \nabla \times \mathbf{U} = \mathbf{J} \times \mathbf{B} - \nabla P \tag{10.46}$$

and then dotting Eq. (10.46) with $\mathbf{U}$ to obtain

$$\rho \frac{\partial}{\partial t} \left(\frac{U^2}{2} \right) + \rho \mathbf{U} \cdot \nabla \left(\frac{U^2}{2} \right) = \mathbf{J} \times \mathbf{B} \cdot \mathbf{U} - \mathbf{U} \cdot \nabla P. \tag{10.47}$$

Multiplying the entire continuity equation by $U^2/2$ and adding the result gives

$$\frac{\partial}{\partial t} \left(\frac{\rho U^2}{2} \right) + \nabla \cdot \left(\frac{\rho U^2}{2} \mathbf{U} \right) = -\mathbf{J} \cdot \mathbf{U} \times \mathbf{B} - \nabla \cdot (P\mathbf{U}) + P \nabla \cdot \mathbf{U}. \tag{10.48}$$

Using Ampère's law, Eq. (10.39), to eliminate $\mathbf{J}$, then Ohm's law, Eq. (10.37), to eliminate $\mathbf{U} \times \mathbf{B}$, and the pressure evolution equation, Eq. (10.45), to eliminate $P \nabla \cdot \mathbf{U}$, this energy equation becomes

$$\frac{\partial}{\partial t} \left(\frac{\rho U^2}{2} \right) + \nabla \cdot \left(\frac{\rho U^2}{2} \mathbf{U} \right) = \frac{(\nabla \times \mathbf{B})}{\mu_0} \cdot \mathbf{E} - \nabla \cdot (P\mathbf{U}) - \frac{1}{(\gamma - 1)} \left(\frac{\partial P}{\partial t} + \nabla \cdot (P\mathbf{U}) \right). \tag{10.49}$$

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Finally, using the vector identity

$$\begin{aligned}\nabla \cdot (\mathbf{E} \times \mathbf{B}) &= \mathbf{B} \cdot \nabla \times \mathbf{E} - \mathbf{E} \cdot \nabla \times \mathbf{B} \\ &= -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \nabla \times \mathbf{B}\end{aligned} \tag{10.50}$$

Eq. (10.49) can be written as

$$\frac{\partial}{\partial t}\left(\frac{\rho U^2}{2} + \frac{P}{\gamma - 1} + \frac{B^2}{2\mu_0}\right) + \nabla \cdot \left(\rho \frac{U^2}{2}\mathbf{U} + \frac{\mathbf{E} \times \mathbf{B}}{\mu_0} + \frac{\gamma}{\gamma - 1} P\mathbf{U}\right) = 0. \tag{10.51}$$

This is a conservation equation relating energy density and energy flux. The time derivative operates on the magnetofluid energy density and the divergence operates on the energy flux. The energy density is comprised of kinetic energy density $\rho U^2/2$, thermal energy density $P/(\gamma - 1)$, and magnetic energy $B^2/2\mu_0$. The energy flux is comprised of convection of kinetic energy $\rho U^2 \mathbf{U}/2$, the Poynting vector $\mathbf{E} \times \mathbf{B}/\mu_0$, and convection of thermal energy density $\gamma P\mathbf{U}/(\gamma - 1)$.

The pressure and density both vanish at the magnetofluid surface, but the Poynting flux $\mathbf{E} \times \mathbf{B}$ can be finite. However, if the tangential electric field vanishes at the surface, then the Poynting flux normal to the surface will be zero. This situation occurs if the magnetofluid is bounded by a perfectly conducting wall with no vacuum region between the magnetofluid and the wall. On the other hand, if a vacuum region bounds the magnetofluid, then a tangential electric field can exist at the vacuum–magnetofluid interface and allow a Poynting flux normal to the surface. Energy could then flow back and forth between the magnetofluid and the vacuum region. For example, if the magnetofluid were to move towards the wall thereby reducing the volume of the vacuum region, magnetic field in the vacuum region would be compressed and so the vacuum magnetic field energy would increase. This flow of energy into the vacuum region would require a Poynting flux from the magnetofluid into the vacuum region.

Let us now consider the energy properties of the vacuum region between the magnetofluid and a perfectly conducting wall. The equations characterizing the vacuum region are Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{10.52}$$

and Ampère's law

$$\nabla \times \mathbf{B} = 0. \tag{10.53}$$

Dotting Faraday's law with $\mathbf{B}$, Ampère's law with $\mathbf{E}$, and subtracting gives

$$\frac{\partial}{\partial t}\left(\frac{B^2}{2\mu_0}\right) + \nabla \cdot \left(\frac{\mathbf{E} \times \mathbf{B}}{\mu_0}\right) = 0, \tag{10.54}$$

which is just the limit of Eq. (10.51) for zero density and zero pressure. Thus, Eq. (10.51) characterizes not only the magnetofluid, but also the surrounding vacuum region. As mentioned earlier, the Poynting flux is the means by which electromagnetic energy flows between the magnetofluid and the vacuum region.

If the vacuum region is bounded by a conducting wall, then the tangential electric field must vanish on the wall. The integral of the energy equation over the volume of both the magnetofluid and the surrounding vacuum region then becomes

$$\frac{\partial}{\partial t}\int d^3r\left(\frac{\rho U^2}{2}+\frac{P}{\gamma-1}+\frac{B^2}{2\mu_0}\right)=0, \tag{10.55}$$

since on the wall $ds\cdot\mathbf{E}\times\mathbf{B}=0$, where $d\mathbf{s}$ is a surface element of the wall. If the wall is brought right up to the magnetofluid so there is no vacuum region, then Eq. (10.55) will also result, with the additional stipulation that the normal component of the velocity vanishes at the wall (i.e., the wall is impermeable).

A system consisting of a magnetofluid surrounded by a vacuum region enclosed by an impermeable perfectly conducting wall will therefore have its total internal energy conserved, that is

$$\int_{V_{mf}} d^3r\left(\frac{\rho U^2}{2}+\frac{P}{\gamma-1}+\frac{B^2}{2\mu_0}\right)+\int_{V_{vac}} d^3r\frac{B^2}{2\mu_0}=const., \tag{10.56}$$

where V_{mf} is the volume of the magnetofluid and V_{vac} is the volume of the vacuum region between the magnetofluid and the wall.

This total system energy can be split into a kinetic energy term

$$T=\int d^3r\frac{\rho U^2}{2} \tag{10.57}$$

and a potential energy term

$$W=\int_V d^3r\left(\frac{P}{\gamma-1}+\frac{B^2}{2\mu_0}\right) \tag{10.58}$$

so that

$$T+W=\mathcal{E}, \tag{10.59}$$

where the total energy $\mathcal{E}$ is a constant. Here V includes the volume of both the magnetofluid and any vacuum region between the magnetofluid and the wall.

We will now consider a static equilibrium (i.e., an equilibrium with $\mathbf{U}_0=0$) so that

$$0=\mathbf{J}_0\times\mathbf{B}_0-\nabla P_0. \tag{10.60}$$

Thus ∇P_0 is normal to the surface defined by the $\mathbf{J}_0$ and $\mathbf{B}_0$ vectors. Furthermore, dotting Eq. (10.60) with $\mathbf{B}_0$ and then with $\mathbf{J}_0$ gives the two relations

$$\mathbf{B}_0\cdot\nabla P_0=0, \quad \mathbf{J}_0\cdot\nabla P_0=0, \tag{10.61}$$

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which indicate that the pressure is constant along both a magnetic field line and a line defined by the current density vector $\mathbf{J}_0$. Since the analysis of the Grad–Shafranov equation in Section 9.8.3 showed that $\mathbf{B}_0$ and $\mathbf{J}_0$ typically lie in a magnetic surface, the pressure is uniform on the magnetic surface, and the pressure gradient is normal to the magnetic surface.

10.4.2 Self-adjointness of potential energy as a consequence of energy integral

Static equilibrium means $\mathbf{U}_0 = 0$ and so Equation (10.57) indicates $T_0 = 0$ in static equilibrium. Thus, all internal energy in equilibrium must exist in the form of stored potential energy, i.e.,

$$W_0 = \mathcal{E}. \tag{10.62}$$

It is now supposed that random thermal noise causes small motions of order ϵ to develop at each point in the magnetofluid, i.e., at each point there exists a first-order velocity

$$\mathbf{U}_1 \sim \epsilon. \tag{10.63}$$

The displacement of a fluid element is obtained by time integration to be

$$\boldsymbol{\xi} = \int_0^t \mathbf{U}_1 \mathrm{d}t' \tag{10.64}$$

and so $\boldsymbol{\xi}$ is also of order ϵ. Because ϵ is small, any spatial dependence of $\mathbf{U}_1$ can be ignored when evaluating the time integral, i.e., terms of order $\boldsymbol{\xi} \cdot \nabla \mathbf{U}_1$ can be ignored since these are of order ϵ^2.

The kinetic energy associated with the mode is

$$\delta T = \int \mathrm{d}^3 r \frac{\rho_0 U_1^2}{2}, \tag{10.65}$$

which clearly is of order ϵ^2. Since total energy is conserved it is necessary to have

$$\delta T + \delta W = 0 \tag{10.66}$$

leading to the important conclusion that the perturbed potential energy δW must also be of order ϵ^2.

The perturbed magnetic field and pressure can be found to first-order in ϵ by integrating Eqs. (10.42) and (10.44) respectively to obtain

$$\mathbf{B}_1 = \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_0) \tag{10.67}$$

and

$$P_1 = -\boldsymbol{\xi} \cdot \nabla P_0 - \gamma P_0 \nabla \cdot \boldsymbol{\xi}; \tag{10.68}$$

these relations show that both $\mathbf{B}_1$ and P_1 are linear functions of $\boldsymbol{\xi}$. However, since it was just shown that δW scales as ϵ^2, only terms that are second-order in $\boldsymbol{\xi}$ can contribute to δW, and so all first-order terms must average to zero when performing the volume integration required to evaluate δW. To specify that δW depends only to second-order on $\boldsymbol{\xi}$, we write

$$\delta W = \delta W(\boldsymbol{\xi}, \boldsymbol{\xi}), \tag{10.69}$$

where the double argument means that δW is a bilinear function, i.e., $\delta W(a\boldsymbol{\xi}, b\boldsymbol{\eta}) = ab\delta W(\boldsymbol{\xi}, \boldsymbol{\eta})$ for arbitrary $\boldsymbol{\xi}$, $\boldsymbol{\eta}$. The time derivative of δW is thus

$$\delta\dot{W} = \delta W(\dot{\boldsymbol{\xi}}, \boldsymbol{\xi}) + \delta W(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}). \tag{10.70}$$

Since $\dot{\boldsymbol{\xi}}$ is algebraically independent of $\boldsymbol{\xi}$, Eq. (10.70) implies $\delta\dot{W}$ is self-adjoint (i.e., $\delta\dot{W}$ is invariant when its two arguments are interchanged). Self-adjointness is a direct consequence of the existence of an energy integral.

10.4.3 Formal solution for perturbed potential energy

The self-adjointness property can be exploited by explicitly calculating the time derivative of the perturbed potential energy. This is done using the linearized equation of motion

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \mathbf{F}_1, \tag{10.71}$$

where

$$\mathbf{F}_1 = \mathbf{J}_0 \times \mathbf{B}_1 + \mathbf{J}_1 \times \mathbf{B}_0 - \nabla P_1 \tag{10.72}$$

results from linear operations on $\boldsymbol{\xi}$, since $\mathbf{B}_1$, $\mu_0\mathbf{J}_1 = \nabla \times \mathbf{B}_1$, and P_1 all result from linear operations on $\boldsymbol{\xi}$.

Multiplying Eq. (10.71) by the perturbed velocity $\dot{\boldsymbol{\xi}}$ and integrating over the volume of the magnetofluid and vacuum region gives

$$\int \mathrm{d}^3 r \rho_0 \frac{\partial}{\partial t}\left(\frac{\dot{\xi}^2}{2}\right) = \int \mathrm{d}^3 r \dot{\boldsymbol{\xi}} \cdot \mathbf{F}_1(\boldsymbol{\xi}). \tag{10.73}$$

Although the integration includes the vacuum region, both the right- and left-hand sides of this equation vanish in the vacuum region because no density, current, or pressure exist there.

The left-hand side of Eq. (10.73) is just the time derivative of the kinetic energy $\delta\dot{T}$. Since $\delta\dot{T} + \delta\dot{W} = 0$, it is seen that

$$\delta\dot{W} = -\int \mathrm{d}^3 r \dot{\boldsymbol{\xi}} \cdot \mathbf{F}_1(\boldsymbol{\xi}). \tag{10.74}$$

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However, because $\delta\dot{W}$ was shown to be self-adjoint, Eq. (10.74) can also be written in an alternative form as $\delta\dot{W} = -\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\dot{\boldsymbol{\xi}})$. Combining Eq. (10.74) and its alternative form provides an integrable form for $\delta\dot{W}$, namely

$$\begin{aligned}\delta\dot{W} &= -\frac{1}{2}\left(\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\dot{\boldsymbol{\xi}}) + \int \mathrm{d}^3 r \dot{\boldsymbol{\xi}} \cdot \mathbf{F}_1(\boldsymbol{\xi})\right) \\ &= -\frac{1}{2}\frac{\partial}{\partial t}\left(\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\boldsymbol{\xi})\right). \end{aligned} \tag{10.75}$$

Integrating the above equation with respect to time gives the change in system potential energy associated with an arbitrary fluid displacement, namely

$$\delta W = -\frac{1}{2}\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\boldsymbol{\xi}). \tag{10.76}$$

Standard techniques of normal mode analysis can now be invoked and used to provide some restrictions on the form of any dynamical behavior. In particular, by assuming that the displacement has the form

$$\boldsymbol{\xi} = \mathrm{Re}\left(\bar{\boldsymbol{\xi}} e^{-\mathrm{i}\omega t}\right), \tag{10.77}$$

the equation of motion can be written as

$$-\omega^2 \rho \bar{\boldsymbol{\xi}} = \mathbf{F}_1(\bar{\boldsymbol{\xi}}). \tag{10.78}$$

Then, multiplication by $\bar{\boldsymbol{\xi}}^*$ and integrating over volume gives

$$-\omega^2 \int \mathrm{d}^3 r \rho |\bar{\xi}|^2 = \int \mathrm{d}^3 r \bar{\boldsymbol{\xi}}^* \cdot \mathbf{F}_1(\bar{\boldsymbol{\xi}}). \tag{10.79}$$

The discussion of Eq. (10.70) showed that $\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\boldsymbol{\eta})$ is self-adjoint when $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$ are arbitrary *real* variables. However, the linearity of $\mathbf{F}_1$ means that $\int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{F}_1(\boldsymbol{\eta})$ is self-adjoint even if $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$ are complex since $\int \mathrm{d}^3 r\ (\boldsymbol{\xi}_r + i\boldsymbol{\xi}_i) \cdot \mathbf{F}_1(\boldsymbol{\eta}_r + i\boldsymbol{\eta}_i) = \int \mathrm{d}^3 r\ (\boldsymbol{\xi}_r + i\boldsymbol{\xi}_i) \cdot (\mathbf{F}_1(\boldsymbol{\eta}_r) + i\mathbf{F}_1(\boldsymbol{\eta}_r)) = \int \mathrm{d}^3 r\ (\boldsymbol{\eta}_r + i\boldsymbol{\eta}_i) \cdot (\mathbf{F}_1(\boldsymbol{\xi}_r + i\boldsymbol{\xi}_i)))$. Equation (10.79) can therefore be recast as

$$-\omega^2 \int \mathrm{d}^3 r \rho |\bar{\xi}|^2 = \frac{1}{2}\left[\int \mathrm{d}^3 r \bar{\boldsymbol{\xi}}^* \cdot \mathbf{F}_1(\bar{\boldsymbol{\xi}}) + \int \mathrm{d}^3 r \bar{\boldsymbol{\xi}} \cdot \mathbf{F}_1(\bar{\boldsymbol{\xi}}^*)\right], \tag{10.80}$$

which shows that ω^2 must be pure real. Negative ω^2 corresponds to instability.

Equation (10.78) can be considered as an eigenvalue problem where ω^2 is the eigenvalue and $\boldsymbol{\xi}$ is the eigenvector. As in the usual linear algebra sense, eigenvectors having different eigenvalues are orthogonal and so, in principle, a basis set of normalized orthogonal eigenvectors $\{\boldsymbol{\zeta}_m\}$ can be constructed where

$$\mathbf{F}_1(\boldsymbol{\zeta}_m) = -\omega_m^2 \rho \boldsymbol{\zeta}_m. \tag{10.81}$$

Thus, any arbitrary displacement can be expressed as a suitably weighted sum of eigenvectors,

$$\boldsymbol{\xi} = \sum_m \alpha_m \boldsymbol{\zeta}_m. \tag{10.82}$$

The orthogonality of the basis set gives the relation

$$\int \mathrm{d}^3 r \boldsymbol{\zeta}_m \cdot \mathbf{F}(\boldsymbol{\zeta}_n) = -\omega_m^2 \int \mathrm{d}^3 r \rho \boldsymbol{\zeta}_m \cdot \boldsymbol{\zeta}_n = 0 \text{ if } m \neq n. \tag{10.83}$$

Instability will result if there exists some perturbation that makes δW negative. Conversely, if all possible perturbations result in positive δW for a given equilibrium, then the equilibrium is MHD stable.

10.4.4 Evaluation of δW

Since $\mathbf{F}_1$ consists of three terms involving $\mathbf{J}_{0,}\mathbf{B}_1$, and P_1 respectively, δW can be decomposed into three terms,

$$\delta W = \delta W_{J_0} + \delta W_{B_1} + \delta W_{P_1}. \tag{10.84}$$

Using Eqs. (10.67), (10.68), and (10.72) in Eq. (10.76) shows that these terms are

$$\delta W_{J_0} = -\frac{1}{2} \int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \mathbf{J}_0 \times \mathbf{B}_1, \tag{10.85}$$

$$\delta W_{B_1} = -\frac{1}{2\mu_0} \int \mathrm{d}^3 r \boldsymbol{\xi} \cdot (\nabla \times \mathbf{B}_1) \times \mathbf{B}_0, \tag{10.86}$$

$$\delta W_{P1} = \frac{1}{2} \int \mathrm{d}^3 r \boldsymbol{\xi} \cdot \nabla P_1. \tag{10.87}$$

Although the above three right-hand sides are in principle integrated over the volumes of both the magnetofluid and the vacuum regions, in fact, the integrands vanish in the vacuum region for all three cases and so the integration is effectively over the magnetofluid volume only. The second two integrals can be simplified using the vector identities $\nabla \cdot (\mathbf{C} \times \mathbf{D}) = \mathbf{D} \cdot \nabla \times \mathbf{C} - \mathbf{C} \cdot \nabla \times \mathbf{D}$ and $\nabla \cdot (f\mathbf{a}) = \mathbf{a} \cdot \nabla f + f \nabla \cdot \mathbf{a}$ to obtain

$$\begin{aligned} \delta W_{B_1} &= \frac{1}{2\mu_0} \int_{V_{mf}} \mathrm{d}^3 r \, (\boldsymbol{\xi} \times \mathbf{B}_0) \cdot \nabla \times \mathbf{B}_1 \\ &= \frac{1}{2\mu_0} \int_{V_{mf}} \mathrm{d}^3 r B_1^2 + \frac{1}{2\mu_0} \int_{S_{mf}} \mathrm{d}\mathbf{s} \cdot \mathbf{B}_1 \times (\boldsymbol{\xi} \times \mathbf{B}_0) \end{aligned} \tag{10.88}$$

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and

$$\delta W_{P_1} = -\frac{1}{2}\int_{V_{mf}} \mathrm{d}^3r\, \boldsymbol{\xi}\cdot\nabla\left[\boldsymbol{\xi}\cdot\nabla P_0 + \gamma P_0 \nabla\cdot\boldsymbol{\xi}\right]$$

$$= \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3r\left[-\boldsymbol{\xi}\cdot\nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right) + \gamma P_0\left(\nabla\cdot\boldsymbol{\xi}\right)^2\right] - \frac{1}{2}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}\gamma P_0\nabla\cdot\boldsymbol{\xi}. \tag{10.89}$$

Since both $\mathbf{J}\times\mathbf{B}$ and ∇P are perpendicular to $\mathbf{B}$ at the surface of the magnetofluid, the force acting on the magnetofluid at the surface is perpendicular to $\mathbf{B}$ and so the displacement $\boldsymbol{\xi}$ of the magnetofluid at the surface is perpendicular to $\mathbf{B}$, i.e., at the surface $\boldsymbol{\xi} = \boldsymbol{\xi}_\perp$. The surface integral in Eq. (10.88) can be expanded

$$\frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\mathbf{B}_1\times\left(\boldsymbol{\xi}\times\mathbf{B}_0\right) = \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp\mathbf{B}_0\cdot\mathbf{B}_1, \tag{10.90}$$

since $\mathrm{d}\mathbf{s}\cdot\mathbf{B}_0 = 0$ on the magnetofluid surface. This latter condition is true because the magnetic field was initially tangential to the magnetofluid surface (i.e., $\mathbf{J}_0\times\mathbf{B}_0 = \nabla P_0$ implies ∇P_0 is perpendicular to $\mathbf{B}_0$) and must remain so since the field is frozen into the magnetofluid.

Recombining and reordering these separate contributions gives

$$\delta W = \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3r\left\{\gamma P_0\left(\nabla\cdot\boldsymbol{\xi}\right)^2 + \frac{B_1^2}{\mu_0} - \boldsymbol{\xi}\cdot\left[\mathbf{J}_0\times\mathbf{B}_1 + \nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right)\right]\right\}$$

$$+ \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp\left[\mathbf{B}_0\cdot\mathbf{B}_1 - \mu_0\gamma P_0\nabla\cdot\boldsymbol{\xi}\right]. \tag{10.91}$$

The substitution $\mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi} = \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp$ has been made for the pressure contribution to the surface term on the grounds that $\mathrm{d}\mathbf{s}$ must be perpendicular to $\mathbf{B}_0$. Further simplification is obtained by considering the dot product with $\mathbf{B}_0$ of the term in square brackets in Eq. (10.91), namely

$$\begin{aligned}
\mathbf{B}_0\cdot\left[\mathbf{J}_0\times\mathbf{B}_1 + \nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right)\right] &= -\nabla P_0\cdot\mathbf{B}_1 + \mathbf{B}_0\cdot\nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right) \\
&= -\nabla P_0\cdot\nabla\times\left(\boldsymbol{\xi}\times\mathbf{B}_0\right) + \mathbf{B}_0\cdot\nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right) \\
&= \nabla\cdot\left\{\nabla P_0\times\left(\boldsymbol{\xi}\times\mathbf{B}_0\right) + \left(\boldsymbol{\xi}\cdot\nabla P_0\right)\mathbf{B}_0\right\} \\
&= \nabla\cdot\left[\boldsymbol{\xi}\mathbf{B}_0\cdot\nabla P_0\right] \\
&= 0,
\end{aligned} \tag{10.92}$$

where Eq. (10.61) has been invoked to obtain the last line. Thus, $\boldsymbol{\xi}\cdot\left[\mathbf{J}_0\times\mathbf{B}_1 + \nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right)\right] = \boldsymbol{\xi}_\perp\cdot\left[\mathbf{J}_0\times\mathbf{B}_1 + \nabla\left(\boldsymbol{\xi}\cdot\nabla P_0\right)\right]$ since Eq. (10.92) shows that the factor in square brackets has no component parallel to the equilibrium magnetic field.

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The potential energy variation δW is now decomposed into its magnetofluid volume and surface components,

$$\delta W_{F'} = \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma P_0 (\nabla\cdot\boldsymbol{\xi})^2 + \dfrac{B_{1\perp}^2}{\mu_0} + \dfrac{B_{1\parallel}^2}{\mu_0} - \\ \boldsymbol{\xi}_\perp\cdot\mathbf{J}_0\times\mathbf{B}_1 - \boldsymbol{\xi}_\perp\cdot\nabla_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0) \end{array} \right\} \tag{10.93}$$

and

$$\delta W_{S'} = \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp (\mathbf{B}_0\cdot\mathbf{B}_1 - \mu_0\gamma P_0\nabla\cdot\boldsymbol{\xi}). \tag{10.94}$$

At this point it is useful to examine the parallel and perpendicular components of $\mathbf{B}_1$. Finite $\mathbf{B}_{1\perp}$ corresponds to changing the curvature of field lines (twanging or plucking) whereas finite $\mathbf{B}_{1\parallel}$ corresponds to compressing or rarefying the density of field lines. The equilibrium force balance can be written as

$$\mu_0\nabla P_0 = -\nabla_\perp\frac{B_0^2}{2} + \boldsymbol{\kappa}B_0^2, \tag{10.95}$$

where $\boldsymbol{\kappa} = \hat{B}_0\cdot\nabla\hat{B}_0$ is a measure of the local curvature of the equilibrium magnetic field (see discussion regarding Eq. (9.14)). Applying the vector identity $\nabla(\mathbf{C}\cdot\mathbf{D}) = \mathbf{C}\cdot\nabla\mathbf{D} + \mathbf{D}\cdot\nabla\mathbf{C} + \mathbf{D}\times\nabla\times\mathbf{C} + \mathbf{C}\times\nabla\times\mathbf{D}$ provides an alternate form for $\boldsymbol{\kappa}$, namely

$$\boldsymbol{\kappa} = -\hat{B}_0\times\nabla\times\hat{B}_0, \tag{10.96}$$

which will now be used in the evaluation of $B_{1\parallel}$, the parallel component of the perturbed magnetic field. From Eq. (10.67) it is seen that

$$\begin{aligned} B_{1\parallel} &= \hat{B}_0\cdot\nabla\times(\boldsymbol{\xi}_\perp\times\mathbf{B}_0) \\ &= (\boldsymbol{\xi}_\perp\times\mathbf{B}_0)\cdot\nabla\times\hat{B}_0 + \nabla\cdot\left[(\boldsymbol{\xi}_\perp\times\mathbf{B}_0)\times\hat{B}_0\right] \\ &= -B_0\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} - \nabla\cdot(B_0\boldsymbol{\xi}_\perp) \\ &= -B_0\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} - B_0(\nabla\cdot\boldsymbol{\xi}_\perp) - \frac{\boldsymbol{\xi}_\perp}{B_0}\cdot\nabla\frac{B_0^2}{2} \\ &= -B_0[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + \nabla\cdot\boldsymbol{\xi}_\perp] + \frac{\mu_0}{B_0}\boldsymbol{\xi}_\perp\cdot\nabla P_0. \end{aligned} \tag{10.97}$$

The term involving $\mathbf{J}_0$ in Eq. (10.93) can be expanded to give

$$\begin{aligned} \boldsymbol{\xi}_\perp\cdot\mathbf{J}_0\times\mathbf{B}_1 &= \boldsymbol{\xi}_\perp\cdot\mathbf{J}_{0\perp}\times\hat{B}_0 B_{1\parallel} + \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp} \\ &= (\boldsymbol{\xi}_\perp\cdot\nabla P_0)\frac{B_{1\parallel}}{B_0} + \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp}. \end{aligned} \tag{10.98}$$

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Substituting for this term in Eq. (10.93), factoring out one power of $B_{1\parallel}$, and then substituting Eq. (10.97) gives

$$\delta W_{F'} = \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma P_0 (\nabla\cdot\boldsymbol{\xi})^2 + \dfrac{B_{1\perp}^2}{\mu_0} + \\ \dfrac{B_{1\parallel}^2}{\mu_0} - (\boldsymbol{\xi}_\perp\cdot\nabla P_0)\dfrac{B_{1\parallel}}{B_0} - \\ \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp} - \boldsymbol{\xi}_\perp\cdot\nabla_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0) \end{array} \right\}$$

$$= \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma P_0 (\nabla\cdot\boldsymbol{\xi})^2 + \dfrac{B_{1\perp}^2}{\mu_0} + \\ \dfrac{B_{1\parallel}}{\mu_0}\left(B_{1\parallel} - \dfrac{\mu_0\boldsymbol{\xi}_\perp\cdot\nabla P_0}{B_0}\right) - \\ \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp} - \boldsymbol{\xi}_\perp\cdot\nabla_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0) \end{array} \right\}$$

$$= \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma P_0 (\nabla\cdot\boldsymbol{\xi})^2 + \dfrac{B_{1\perp}^2}{\mu_0} + \\ \left(\dfrac{B_0^2}{\mu_0}[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + \nabla\cdot\boldsymbol{\xi}_\perp] - \boldsymbol{\xi}_\perp\cdot\nabla P_0\right)[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + \nabla\cdot\boldsymbol{\xi}_\perp] - \\ \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp} + (\boldsymbol{\xi}_\perp\cdot\nabla P_0)\,\nabla\cdot\boldsymbol{\xi}_\perp - \\ \nabla\cdot[\boldsymbol{\xi}_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0)] \end{array} \right\}$$

$$= \frac{1}{2}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma P_0 (\nabla\cdot\boldsymbol{\xi})^2 + \dfrac{B_{1\perp}^2}{\mu_0} + \dfrac{B_0^2}{\mu_0}[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + \nabla\cdot\boldsymbol{\xi}_\perp]^2 - \\ (\boldsymbol{\xi}_\perp\cdot\nabla P_0)(2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa}) - \boldsymbol{\xi}_\perp\cdot J_{0\parallel}\hat{B}_0\times\mathbf{B}_{1\perp} - \\ \nabla\cdot[\boldsymbol{\xi}_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0)] \end{array} \right\}$$

$$= \frac{1}{2\mu_0}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma\mu_0 P_0 (\nabla\cdot\boldsymbol{\xi})^2 + B_{1\perp}^2 + B_0^2[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + \nabla\cdot\boldsymbol{\xi}_\perp]^2 - \\ 2\mu_0(\boldsymbol{\xi}_\perp\cdot\nabla P_0)(\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa}) + \boldsymbol{\xi}_\perp\times\mathbf{B}_{1\perp}\cdot\left(\mu_0 J_{0\parallel}\hat{B}_0\right) \end{array} \right\}$$
$$-\frac{1}{2}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp(\boldsymbol{\xi}_\perp\cdot\nabla P_0). \qquad (10.99)$$

A new surface term has appeared because of an integration by parts; this new term is absorbed into the previous surface term and the fluid and surface terms are redefined by removing the primes; thus,

$$\delta W_F = \frac{1}{2\mu_0}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma\mu_0 P_0 (\nabla\cdot\boldsymbol{\xi})^2 + B_{1_\perp}^2 + B_0^2[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + (\nabla\cdot\boldsymbol{\xi}_\perp)]^2 - \\ 2\mu_0(\boldsymbol{\xi}_\perp\cdot\nabla P_0)(\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa}) + \boldsymbol{\xi}_\perp\times\mathbf{B}_{1\perp}\cdot\left(\mu_0 J_{0\parallel}\hat{B}_0\right) \end{array} \right\}$$
$$(10.100)$$

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and

$$\delta W_S = \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp \left\{\mathbf{B}_0\cdot\mathbf{B}_1 - \mu_0\left(\gamma P_0 \nabla\cdot\boldsymbol{\xi} + \boldsymbol{\xi}_\perp\cdot\nabla P_0\right)\right\}$$
$$= \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp \left\{\mathbf{B}_0\cdot\mathbf{B}_1 + \mu_0 P_1\right\}, \tag{10.101}$$

where $\mathbf{B}_0\cdot\mathrm{d}\mathbf{s} = 0$ and Eq. (10.68) have been used to simplify the surface integral. The surface integral can be further rearranged by considering the relationship between the vacuum and magnetofluid fields at the perturbed surface. If the equilibrium force balance is integrated over the volume of the small Gaussian pillbox located at the perturbed magnetofluid surface in Fig. 10.5, it is seen that

$$0 = \int_{V_{pillbox,ps}} \mathrm{d}^3 r \nabla\cdot\left[\left(P + \frac{B^2}{2\mu_0}\right)\mathbf{I} - \frac{1}{\mu_0}\mathbf{B}\mathbf{B}\right] = \int_{S,ps} \mathrm{d}s\left[P + \frac{B^2}{2\mu_0}\right]_{magnetofluid}^{vac} \tag{10.102}$$

since $\mathrm{d}\mathbf{s}\cdot\mathbf{B} = 0$ at the perturbed surface. The subscripts *ps* indicate that the volume and surface integrals are at the perturbed surface.

Quantities evaluated at the perturbed surface are of the form

$$f_{pert\,sfc} = f_0 + f_1 + \boldsymbol{\xi}\cdot\nabla f_0; \tag{10.103}$$

i.e., both absolute and convective first-order terms must be included. Since the pillbox extent was arbitrary in Eq. (10.102), the integrand $\left[P + B^2/2\mu_0\right]_{magnetofluid}^{vac}$ must vanish at each point on the perturbed surface, giving the relation

$$\left(P_1 + \frac{\mathbf{B}_0\cdot\mathbf{B}_1}{\mu_0}\right)_{\substack{magneto\\fluid}} + \boldsymbol{\xi}\cdot\nabla\left(P_0 + \frac{B_0^2}{2\mu_0}\right)_{\substack{magneto\\fluid}} = \left(\frac{\mathbf{B}_0\cdot\mathbf{B}_1}{\mu_0}\right)_{vac} + \boldsymbol{\xi}\cdot\nabla\left(\frac{B_0^2}{2\mu_0}\right)_{vac}, \tag{10.104}$$

which can be rewritten as

$$\left(P_1 + \frac{\mathbf{B}_0\cdot\mathbf{B}_1}{\mu_0}\right)_{magnetofluid} = \boldsymbol{\xi}\cdot\nabla\left[P_0 + \frac{B_0^2}{2\mu_0}\right]_{magnetofluid}^{vac} + \left(\frac{\mathbf{B}_0\cdot\mathbf{B}_1}{\mu_0}\right)_{vac}. \tag{10.105}$$

Thus, the surface integral Eq. (10.101) can be rewritten as

$$\delta W_S = \frac{1}{2}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp\boldsymbol{\xi}_\perp\cdot\nabla\left[P_0 + \frac{B_0^2}{2\mu_0}\right]_{magnetofluid}^{vac} + \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp\left(\mathbf{B}_0\cdot\mathbf{B}_1\right)_{vac}. \tag{10.106}$$

The volume integral in Eq. (10.100) is over the magnetofluid volume and the direction of $\mathrm{d}\mathbf{s}$ in Eq. (10.106) is outwards from the magnetofluid volume. The energy stored in the vacuum region between the magnetofluid and wall is

$$\begin{aligned}\delta W_{vac} &= \frac{1}{2\mu_0}\int_{V_{vac}} \mathrm{d}^3 r B_{1v}^2 \\ &= \frac{1}{2\mu_0}\int_{V_{vac}} \mathrm{d}^3 r \mathbf{B}_{1v}\cdot\nabla\times\mathbf{A}_{1v} \\ &= \frac{1}{2\mu_0}\int_{V_{vac}} \mathrm{d}^3 r \nabla\cdot(\mathbf{A}_{1v}\times\mathbf{B}_{1v}) \\ &= \frac{1}{2\mu_0}\int_{S_{vac}} \mathrm{d}\mathbf{s}\cdot(\mathbf{A}_{1v}\times\mathbf{B}_{1v}),\end{aligned} \tag{10.107}$$

where $\mathrm{d}\mathbf{s}$ points *out* from the vacuum region and $\nabla\times\mathbf{B}_{1v}=0$ has been used when integrating by parts. At the conducting wall (which might be at infinity) the integrand vanishes, since it contains the factor $\mathrm{d}\mathbf{s}\times\mathbf{A}_{1v}$, which is proportional to the tangential electric field, a quantity that vanishes at a conductor. Thus, only the surface integral over the magnetofluid–vacuum interface remains. Since an element of surface $\mathrm{d}\mathbf{s}$ pointing out of the vacuum points into the magnetofluid, on using $\mathrm{d}\mathbf{s}$ to mean out of the magnetofluid (as before), Eq. (10.107) becomes

$$\delta W_{vac} = -\frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot(\mathbf{A}_{1v}\times\mathbf{B}_{1v}), \tag{10.108}$$

where S_{mf} has been used instead of S_{vac} to indicate that $\mathrm{d}\mathbf{s}$ points out of the magnetofluid.

The tangential component of the electric field must be continuous at the magnetofluid–vacuum interface. This field is not necessarily zero. The electric field inside the magnetofluid is determined by Ohm's law, Eq. (10.37). Thus, the electric field on the magnetofluid side of the magnetofluid–vacuum interface is

$$\mathbf{E}_{1p} = -\mathbf{v}_1\times\mathbf{B}_0 \tag{10.109}$$

while on the vacuum side of this interface the electric field is simply

$$\mathbf{E}_{1v} = -\frac{\partial\mathbf{A}_{1v}}{\partial t}, \tag{10.110}$$

where $\mathbf{A}_{1v}$ is the vacuum vector potential. Defining the surface normal unit vector $\hat{n}$ and integrating both electric fields with respect to time, the condition that the tangential electric field is continuous is seen to be

$$\hat{n}\times\mathbf{A}_{1v} = \hat{n}\times(\boldsymbol{\xi}\times\mathbf{B}_0) = -\hat{n}\cdot\boldsymbol{\xi}\mathbf{B}_0, \tag{10.111}$$

where $\hat{n}\cdot\mathbf{B}_0=0$ has been used. Thus, the second surface integral in Eq. (10.106) can be written as

$$\frac{1}{2\mu_0}\int_S \mathrm{d}s\hat{n}\cdot\boldsymbol{\xi}_\perp\mathbf{B}_0\cdot\mathbf{B}_{1v} = -\frac{1}{2\mu_0}\int_S \mathrm{d}s\hat{n}\times\mathbf{A}_{1v}\cdot\mathbf{B}_{1v} = \delta W_{vac}. \tag{10.112}$$

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These results are now summarized. The perturbed potential energy can be expressed as

$$\delta W = \delta W_F + \delta W_{int} + \delta W_{vac}, \tag{10.113}$$

where the contribution from the fluid volume interior is

$$\delta W_F = \frac{1}{2\mu_0}\int_{V_{mf}} \mathrm{d}^3 r \left\{ \begin{array}{l} \gamma\mu_0 P_0 (\nabla\cdot\boldsymbol{\xi})^2 + B_{1_\perp}^2 + B_0^2\left[2\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa} + (\nabla\cdot\boldsymbol{\xi}_\perp)\right]^2 \\ +\boldsymbol{\xi}_\perp\times\mathbf{B}_{1\perp}\cdot\hat{B}_0\mu_0 J_{0\|} - 2\mu_0(\boldsymbol{\xi}_\perp\cdot\boldsymbol{\kappa})(\boldsymbol{\xi}_\perp\cdot\nabla P_0) \end{array}\right\}, \tag{10.114}$$

the contribution from the vacuum–magnetofluid interface is

$$\delta W_{int} = \frac{1}{2\mu_0}\int_{S_{mf}} \mathrm{d}\mathbf{s}\cdot\boldsymbol{\xi}_\perp\boldsymbol{\xi}_\perp\cdot\nabla\left[\mu_0 P_0 + \frac{B_0^2}{2}\right]_{magnetofluid}^{vac}, \tag{10.115}$$

and the contribution from the vacuum region is

$$\delta W_{vac} = \frac{1}{2\mu_0}\int_{vac} \mathrm{d}^3 r\ B_{1v}^2. \tag{10.116}$$

10.5 Discussion of the energy principle

The terms in the integrands of δW_F, δW_{int}, and δW_{vac} are of two types: those that are positive-definite and those that are not. Positive-definite terms always increase δW and are therefore stabilizing whereas terms that could be negative are potentially destabilizing. Consideration of δW_F in particular shows that magnetic perturbations interior to the magnetofluid and perpendicular to the equilibrium field are always stabilizing, since these perturbations appear in the form $B_{1_\perp}^2$. It is also seen that a positive-definite term $\sim(\nabla\cdot\boldsymbol{\xi})^2$ exists indicating that a compressible magnetofluid is always more stable than an otherwise identical incompressible magnetofluid. Hence, incompressible instabilities are more violent than compressible ones. It is also seen that two types of destabilizing terms exist. One gives instability if

$$(\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp)(\boldsymbol{\xi}_\perp\cdot\nabla P_0) > 0; \tag{10.117}$$

this is a generalization of the good curvature/bad curvature result obtained in the earlier Rayleigh–Taylor analysis. In the vicinity of the magnetofluid surface all quantities in Eq. (10.117) point in the same direction and so Eq. (10.117) can be written as

$$\boldsymbol{\kappa}\cdot\nabla P_0 > 0 \quad \Longrightarrow \text{instability}, \tag{10.118}$$

which gives instability for bad curvature ($\boldsymbol{\kappa}$ parallel to ∇P_0) and stability for good curvature ($\boldsymbol{\kappa}$ antiparallel to ∇P_0). A Bennett pinch has bad curvature and

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is therefore grossly unstable to Rayleigh–Taylor interchange modes. Instabilities associated with Eq. (10.117) are called pressure-driven instabilities, and are important in plasmas where there is significant energy stored in the pressure (high β where $\beta = 2\mu_0 P_0/B_0^2$).

The other type of destabilizing term depends on the existence of a force-free current (i.e., $J_{0\parallel} \neq 0$) and leads to internal kink instabilities. These sorts of instabilities are called current-driven instabilities (although strictly speaking, only the parallel component of the current is involved).

There also exist instabilities associated with the magnetofluid–vacuum interface. The energy δW_{int} can be thought of as the change in potential energy of a stretched membrane at the magnetofluid–vacuum interface where the stretching force is given by the difference between the vacuum and magnetofluid forces pushing on the membrane surface; instability will occur if $\delta W_{int} < 0$. When investigating these surface instabilities it is convenient to set δW_F to zero by idealizing the plasma to being incompressible, having uniform internal pressure, and no internal currents. Surface instabilities can exist only if the surface can move, and so require a vacuum region between the wall and the plasma. Thus, moving a conducting wall right up to the surface of a conducting plasma prevents surface instabilities. These surface instabilities will be investigated in detail later in this chapter.

10.6 Current-driven instabilities and helicity

We shall now discuss current driven instabilities and show that these are helical in nature and are driven by gradients in $J_{0\parallel}/B_0$. To simplify the analysis $P \to 0$ is assumed so that pressure-driven instabilities can be neglected, since they have already been discussed. On making this simplification, Eq. (10.97) shows that the parallel component of the perturbed magnetic field reduces to

$$B_{1\parallel} = -B_0 \left[2\boldsymbol{\xi}_\perp \cdot \boldsymbol{\kappa} + \nabla \cdot \boldsymbol{\xi}_\perp \right]. \tag{10.119}$$

Thus, we may identify

$$B_1^2 = B_{1\perp}^2 + B_{1\parallel}^2 = \mathbf{B}_{1\perp}^2 + B_0^2 \left[2\boldsymbol{\xi}_\perp \cdot \boldsymbol{\kappa} + \nabla \cdot \boldsymbol{\xi}_\perp \right]^2 \tag{10.120}$$

and so the perturbed potential energy of the magnetofluid volume reduces to

$$\delta W_F = \frac{1}{2\mu_0} \int \mathrm{d}^3 r \left\{ B_1^2 + \boldsymbol{\xi}_\perp \times \mathbf{B}_{1\perp} \cdot \mathbf{B}_0 \frac{\mu_0 J_{0\parallel}}{B_0} \right\}. \tag{10.121}$$

Equation (10.67) shows that the perturbed vector potential can be identified as

$$\mathbf{A}_1 = \boldsymbol{\xi} \times \mathbf{B}_0 \tag{10.122}$$

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so Eq. (10.121) can be recast as

$$\delta W_F = \frac{1}{2\mu_0}\int \mathrm{d}^3 r \left\{ B_1^2 - \mathbf{A}_1\cdot\mathbf{B}_1 \frac{\mu_0 J_{0\parallel}}{B_0} \right\}. \tag{10.123}$$

We now show that finite $\mathbf{A}_1\cdot\mathbf{B}_1$ corresponds to a helical perturbation. Consider the simplest situation where $\mathbf{A}_1\cdot\mathbf{B}_1$ is simply a constant and define a local Cartesian coordinate system with z axis parallel to the local $\mathbf{B}_0$. Equation (10.122) shows that $\mathbf{A}_1 = A_{1x}\hat{x} + A_{1y}\hat{y}$ so

$$\mathbf{A}_1\cdot\mathbf{B}_1 = -A_{1x}\frac{\partial A_{1y}}{\partial z} + A_{1y}\frac{\partial A_{1x}}{\partial z}. \tag{10.124}$$

Suppose both components of $\mathbf{A}_1$ are non-trivial functions of z and, in particular, assume $A_{1x} = \mathrm{Re}\ \mathcal{A}_{1x}\exp(\mathrm{i}kz)$ and $A_{1y} = \mathrm{Re}\ \mathcal{A}_{1y}\exp(\mathrm{i}kz)$. In this case

$$\mathbf{A}_1\cdot\mathbf{B}_1 = \frac{1}{2}\mathrm{Re}\left[-\mathcal{A}_{1x}^*\frac{\partial \mathcal{A}_{1y}}{\partial z} + \mathcal{A}_{1y}^*\frac{\partial \mathcal{A}_{1x}}{\partial z}\right] = -\frac{k}{2}\mathrm{Re}\left[\mathrm{i}\left(\mathcal{A}_{1x}^*\mathcal{A}_{1y} - \mathcal{A}_{1y}^*\mathcal{A}_{1x}\right)\right], \tag{10.125}$$

which can be finite only if $\mathcal{A}_{1x}^*\mathcal{A}_{1y}$ is not pure real. The simplest such case is where $\mathcal{A}_{1y} = \mathrm{i}\mathcal{A}_{1x}$ so

$$\mathbf{A}_1\cdot\mathbf{B}_1 = k|\mathcal{A}_{1x}|^2 \tag{10.126}$$

and

$$\mathbf{A}_1 = \mathrm{Re}\left[\mathcal{A}_{1x}(\hat{x}+\mathrm{i}\hat{y})\exp(\mathrm{i}kz)\right], \tag{10.127}$$

which is a helically polarized field since $A_{1x} \sim \cos kz$ and $A_{1y} \sim \sin kz$.

10.7 Magnetic helicity

Since finite $\mathbf{A}_1\cdot\mathbf{B}_1$ corresponds to the local helical polarization of the perturbed fields, it is reasonable to define $\mathbf{A}\cdot\mathbf{B}$ as the density of magnetic helicity and to define the total magnetic helicity in a volume as

$$K = \int_V \mathrm{d}^3 r \mathbf{A}\cdot\mathbf{B}. \tag{10.128}$$

The question immediately arises whether this definition makes sense, i.e., is it reasonable to define $\mathbf{A}\cdot\mathbf{B}$ as an intensive property and K as an extensive property? An obvious problem is that $\mathbf{A}$ is undefined with respect to a gauge: $\mathbf{A}$ can be redefined to be $\mathbf{A}' = \mathbf{A} + \nabla f$, where f is an arbitrary scalar function since $\mathbf{B} = \nabla\times\mathbf{A} = \nabla\times(\mathbf{A}+\nabla f)$ is unaffected by the choice of gauge. The definition for magnetic helicity would be of little use if K were gauge-dependent because gauge has no physical significance. However, if no magnetic field penetrates the surface S enclosing the volume V, the proposed definition of K is gauge-independent. This

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is because if no magnetic field penetrates the surface then $\mathbf{B}\cdot\mathrm{d}\mathbf{s}=0$ everywhere on the surface. If this is true, then

$$\begin{aligned}\int_V \mathrm{d}^3r\,(\mathbf{A}+\nabla f)\cdot\mathbf{B} &= \int_V \mathrm{d}^3r\mathbf{A}\cdot\mathbf{B}+\int_V \mathrm{d}^3r\nabla\cdot(f\mathbf{B})\\ &= \int_V \mathrm{d}^3r\mathbf{A}\cdot\mathbf{B}+\int_S \mathrm{d}\mathbf{s}\cdot(f\mathbf{B})\\ &= \int_V \mathrm{d}^3r\mathbf{A}\cdot\mathbf{B}\end{aligned} \tag{10.129}$$

and so the total helicity defined by Eq. (10.128) is gauge-independent even though the helicity density $\mathbf{A}\cdot\mathbf{B}$ is gauge-dependent.

Let us consider the situation where there is no vacuum region between the plasma and an impermeable wall. Thus the normal fluid velocity $\mathbf{u}_{1\perp}$ must vanish at the wall. From Ohm's law the component of the perturbed electric field tangential to the wall $\mathbf{E}_{1t}$ must vanish since

$$\mathbf{E}_{1t}=-\mathbf{u}_{1\perp}\times\mathbf{B}_0=0 \tag{10.130}$$

and $\mathbf{B}_0$ lies in the plane of the wall. Thus, an impermeable wall is equivalent to a conducting wall if $\mathbf{B}_0$ lies in the plane of the wall. If the magnetic field initially does not penetrate the wall, i.e., $\mathbf{B}\cdot\mathrm{d}\mathbf{s}=0$ initially, then the field will always remain tangential to the wall and the total helicity K in the volume bounded by the wall will always be a well-defined quantity (i.e., will always be gauge-invariant).

A conservation equation for helicity density can be obtained by combining the time derivative of the magnetic helicity density with Faraday's law. This is seen by direct calculation:

$$\begin{aligned}-\frac{\partial}{\partial t}(\mathbf{A}\cdot\mathbf{B}) &= -\frac{\partial\mathbf{A}}{\partial t}\cdot\mathbf{B}-\mathbf{A}\cdot\frac{\partial\mathbf{B}}{\partial t}\\ &= (\mathbf{E}+\nabla\varphi)\cdot\mathbf{B}+\mathbf{A}\cdot\nabla\times\mathbf{E}\\ &= 2\mathbf{E}\cdot\mathbf{B}+\nabla\cdot(\varphi\mathbf{B}+\mathbf{E}\times\mathbf{A}),\end{aligned} \tag{10.131}$$

where φ is the electrostatic potential. Since the ideal MHD Ohm's law, Eq. (10.37), implies $\mathbf{E}\cdot\mathbf{B}=0$, Eq. (10.131) can be rearranged in the form of a conservation equation,

$$\frac{\partial}{\partial t}(\mathbf{A}\cdot\mathbf{B})+\nabla\cdot(\varphi\mathbf{B}+\mathbf{E}\times\mathbf{A})=0. \tag{10.132}$$

Integration of Eq. (10.132) over the entire magnetofluid volume gives the conservation relation for the total helicity to be

$$\frac{\partial}{\partial t}\left[\int \mathrm{d}^3r\mathbf{A}\cdot\mathbf{B}\right]+\int \mathrm{d}s\hat{n}\cdot(\varphi\mathbf{B}+\mathbf{E}\times\mathbf{A})=0. \tag{10.133}$$

Since both $\mathbf{B}\cdot\hat{n}=0$ and $\mathbf{E}_t=0$ at the wall, this reduces to

$$K=\int \mathrm{d}^3r\mathbf{A}\cdot\mathbf{B}=const.; \tag{10.134}$$

i.e., the total helicity is conserved for an ideal plasma surrounded by a rigid or conducting wall that is not penetrated by any magnetic field lines.

We now recall that $\mathbf{A}_1$ is a linear function of $\boldsymbol{\xi}$ prescribed by Eq. (10.122) and that $\boldsymbol{\xi}$ is assumed to be of order ϵ. Furthermore, we note that the energy principle derivation is unaffected by making the additional assumption that the variation $\delta\mathbf{A}=\mathbf{A}_1$ is exact to all orders in ϵ, i.e., the energy principle is unaffected if we assume no terms $\mathbf{A}_n\sim\epsilon^n$ exist for $n\geq 2$. Let us use these properties to consider the implications of Eq. (10.134) when the magnetic field is perturbed. On writing $\mathbf{B}=\mathbf{B}_0+\mathbf{B}_1$ and $\mathbf{A}=\mathbf{A}_0+\mathbf{A}_1$, helicity conservation as given by Eq. (10.134) implies

$$\begin{aligned}K&=\int \mathrm{d}^3r\mathbf{A}_0\cdot\nabla\times\mathbf{A}_0\\&=\int \mathrm{d}^3r\,(\mathbf{A}_0+\mathbf{A}_1)\cdot\nabla\times(\mathbf{A}_0+\mathbf{A}_1)\\&=\int \mathrm{d}^3r\mathbf{A}_0\cdot\nabla\times\mathbf{A}_0+\int \mathrm{d}^3r\mathbf{A}_1\cdot\nabla\times\mathbf{A}_0\\&\quad+\int \mathrm{d}^3r\mathbf{A}_0\cdot\nabla\times\mathbf{A}_1+\int \mathrm{d}^3r\mathbf{A}_1\cdot\nabla\times\mathbf{A}_1.\end{aligned} \tag{10.135}$$

Thus, to first-order in the perturbation (i.e., to order ϵ),

$$\int \mathrm{d}^3r\mathbf{A}_1\cdot\nabla\times\mathbf{A}_0+\int \mathrm{d}^3r\mathbf{A}_0\cdot\nabla\times\mathbf{A}_1=0 \tag{10.136}$$

and to second-order (i.e., to order ϵ^2, the order relevant to the energy principle),

$$\int \mathrm{d}^3r\mathbf{A}_1\cdot\nabla\times\mathbf{A}_1=\int \mathrm{d}^3r\mathbf{A}_1\cdot\mathbf{B}_1=0. \tag{10.137}$$

This can be compared to Eq. (10.123), the second term of which is the same as Eq. (10.137) except for a factor $-\mu_0 J_{0\|}/B_0$. In general, this factor can be some complicated function of position. However, in the special case where $\mu_0 J_{0\|}/B_0$ does not depend on position, $\mu_0 J_{0\|}/B_0$ may be factored out of the second term in Eq. (10.123) so as to obtain, using Eq. (10.137),

$$\delta W_F=\frac{1}{2\mu_0}\int \mathrm{d}^3r\; B_1^2, \tag{10.138}$$

which is positive-definite and therefore gives absolute stability. Thus, equilibria having spatially uniform $\lambda=\mu_0 J_{0\|}/B_0$ are stable against current-driven modes. Since these equilibria are helical or kinked, they may be considered as being the final relaxed state associated with a kink instability – once a system attains this final state no free energy remains to drive further instability. Since pressure has been assumed to be negligible, Eq. (10.60) implies that the equilibrium current must be parallel to the equilibrium magnetic field, but does not specify the proportionality factor. Thus, if a configuration satisfies

$$\mu_0 \mathbf{J}_0 = \lambda \mathbf{B}_0, \tag{10.139}$$

where λ is spatially uniform, the configuration is stable against any further helical perturbations. This gives rise to the relation

$$\nabla \times \mathbf{B}_0 = \lambda \mathbf{B}_0, \tag{10.140}$$

where λ is spatially uniform. This equation is called a force-free equilibrium and its solutions are helical vector fields, namely fields where the curl of the field is parallel to the field itself. If a field is confined to a plane, then its curl will be normal to the plane and so a field confined to a plane cannot be a solution to Eq. (10.140). The field must be three-dimensional.

To summarize, it has been shown that current-driven instabilities are helical and drive the plasma towards a force-free equilibrium as prescribed by Eq. (10.140). Current-driven instabilities are energized by gradients in $J_{0\|}/B_0$ and become stabilized when $J_{0\|}/B_0$ becomes spatially uniform. Gradients in $J_{0\|}/B_0$ can therefore be considered as free energy available for driving helical modes. When this free energy is depleted, the helical modes are stabilized and the plasma assumes a force-free equilibrium with spatially uniform $J_{0\|}/B_0$.

This tendency to coil up or kink is a means by which the plasma increases its inductance. However, when the plasma coils up into a state satisfying Eq. (10.140) it is in a stable equilibrium. This stable equilibrium represents a local minimum in potential energy. There might be several such local minima, each of which has a different energy level as sketched in Fig. 9.2. As mentioned earlier, this set of discrete energy levels is somewhat analogous to the ground and excited states of a quantum system. Here, the vacuum magnetic field is analogous to the ground state, while the various force-free equilibria (i.e., solutions of Eq. (10.140)) are analogous to the higher energy states.

10.8 Characterization of free-boundary instabilities

The previous section considered internal instabilities of a magnetofluid bounded by a rigid wall with no vacuum region between the magnetofluid and the wall. Let us now consider the other extreme, namely a situation where not only does a vacuum region exist and bound the magnetofluid but, in addition, the location of the vacuum–magnetofluid interface can move. To focus attention on the motion of the magnetofluid–vacuum boundary, the simplest non-trivial configuration will be

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considered, namely a configuration where the interior pressure is both uniform and finite. This means ∇P vanishes in the magnetofluid interior so the entire pressure gradient and therefore the entire $\mathbf{J} \times \mathbf{B}$ force is concentrated in an infinitesimally thin layer at the magnetofluid surface.

Compressibility, i.e., finite $\nabla \cdot \boldsymbol{\xi}$, was shown by the MHD energy principle to be stabilizing. Therefore, if a given system is stable with respect to incompressible modes, it will be even more stable with respect to compressible modes, or equivalently, with respect to modes having finite $\nabla \cdot \boldsymbol{\xi}$. By assuming $\nabla \cdot \boldsymbol{\xi} = 0$ the worst-case instability scenario is therefore considered and, furthermore, the analysis is simplified. Thus, in order to consider the simplest non-trivial instability resulting from motion of the magnetofluid–vacuum boundary, we assume $\nabla \cdot \boldsymbol{\xi} = 0$, cylindrical geometry, and axial uniformity. For example, the Bennett pinch would satisfy these assumptions if all the current were concentrated at the plasma surface. The physical basis of the two main types of current-driven instability, sausage and kink, will first be discussed qualitatively before proceeding with a detailed mathematical discussion.

10.8.1 Qualitative examination of the sausage instability

Consider a Bennett pinch (z-pinch) that is an axially uniform cylindrical plasma with an axial current and an associated azimuthal magnetic field. As discussed above, the pressure is assumed to be (i) uniform in the interior region $r < a$, where a is the plasma radius, and (ii) zero in the exterior region $r > a$. In equilibrium, the radially inward $\mathbf{J} \times \mathbf{B}$ pinch force balances the radially outward force associated with the pressure gradient. Thus $-J_{z0}B_{\theta 0} = \partial P_0/\partial r$ and both sides of this equation are finite only in an infinitesimal surface layer at $r = a$. We now suppose that random thermal noise causes the incompressible plasma to develop small axially periodic constrictions and bulges shown in an exaggerated fashion in Fig. 10.6. The azimuthal magnetic field $B_\theta = \mu_0 I/2\pi r$ at the constrictions is larger than its equilibrium value because $r < a$ at a constriction whereas I is fixed. Thus, at a constriction the pinch force $\sim B_\theta^2/r$ is greater than its equilibrium value and so is stronger than the outward force from the internally uniform pressure. The resulting net force is therefore inwards and so will cause an inwards radial motion, thereby enhancing the constriction. Because the configuration was assumed to be incompressible, the plasma squeezed inwards at constrictions must flow into the interspersed bulges shown in Fig. 10.6. The azimuthal magnetic field at a bulge is weaker than its equilibrium value because $r > a$ at a bulge. Thus, at a bulge the tables are now turned in the competition between outward pressure and inward pinch force. At a bulge the pressure exceeds the weakened pinch force and this force imbalance causes the bulge to increase. Hence, any combination of initial

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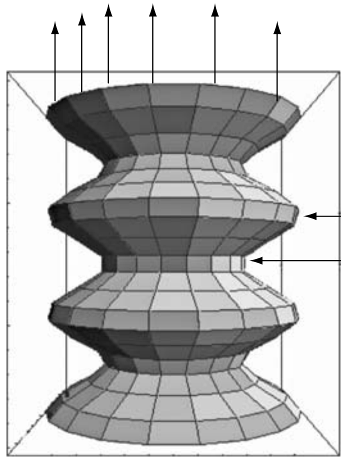

Fig. 10.6 Sausage instability: current is axial, magnetic field is azimuthal.

infinitesimal constrictions and bulges will spontaneously grow in amplitude and so the system is unstable. This behavior is called the "sausage" instability because the end result is a plasma resembling a string of sausages.

Sausage instability can be prevented by either surrounding the plasma with a perfectly conducting wall or by immersing the plasma in a strong axial vacuum magnetic field (i.e., an axial field generated by currents flowing in external solenoid coils that are coaxial with the plasma). If a plasma with embedded axial field attempts to sausage, the axial rippling of the plasma would have to bend the embedded axial field since this axial field is frozen into the plasma. As was shown in Section 9.2 any deformation of a vacuum magnetic field requires work and so the free energy available to drive the sausage instability would have to do work to bend the axial field. If bending the axial magnetic field absorbs more energy than is liberated by the sausaging motion, the plasma is stabilized. The vacuum axial field stabilizes the plasma in a manner analogous to a steel reinforcing rod embedded in concrete. The other method for stabilization, a close-fitting conducting wall, works because image currents induced in the wall by the sausage motion produce a magnetic field that interacts with the plasma current in such a way as to repel the plasma from the wall.

10.8.2 Qualitative examination of the kink instability

The combination of the axial magnetic field B_z proposed in the previous paragraph and the azimuthal magnetic field B_θ produced by the plasma current results in

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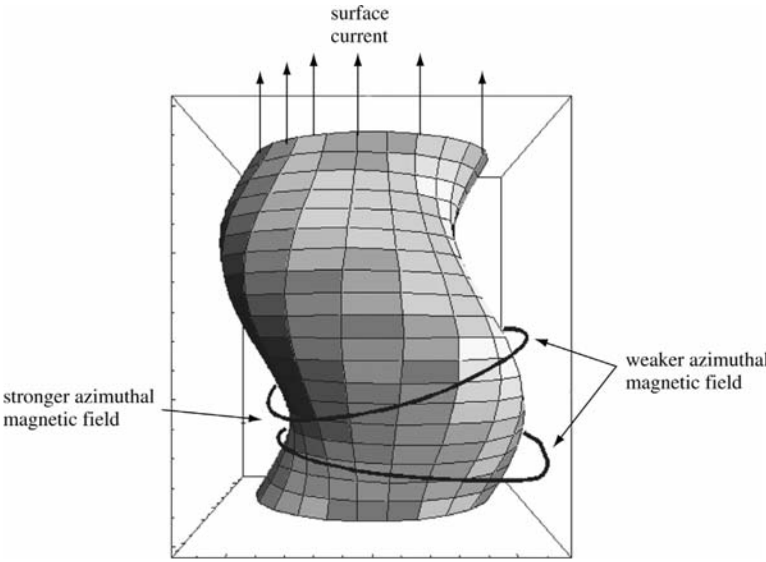

Fig. 10.7 Kink instability can occur when the equilibrium magnetic field is helical.

a helical magnetic field. This helical magnetic field is susceptible to the helical kink instability sketched in Fig. 10.7. At the concave parts of the plasma surface there is a concentration of the azimuthal field resulting in a magnetic pressure that increases the concavity (see lower left of Fig. 10.7). Similarly, at the convex portions of the surface, the azimuthal field is weaker so that the convex bulge will tend to increase (see lower right of Fig. 10.7). This tendency can also be viewed as an example of the hoop force trying to increase the major radius of a segment of curved current. If the externally imposed axial vacuum magnetic field is sufficiently strong, the energy required to deform this vacuum field will exceed the free energy available from the kinking and the instability will be prevented.

10.8.3 Qualitative examination of wall stabilization

Now consider a plasma with no externally produced B_z field but surrounded by a close-fitting perfectly conducting wall. The plasma axial current generates an azimuthal field in the vacuum region between the plasma and the wall. If an instability causes the plasma to move towards the wall, then the vacuum region becomes thinner and the magnetic field in the vacuum region becomes compressed since it cannot penetrate the wall or the plasma. The increased magnetic pressure acts as a restoring force, pushing the plasma back away from the wall. Thus, a close-fitting perfectly conducting wall stabilizes both kinks and sausages.

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10.9 Analysis of free-boundary instabilities

Consider the azimuthally symmetric, axially uniform cylindrical plasma with nominal radius a shown in Fig. 10.8. The plasma is subdivided into two concentric regions consisting of (i) an interior region $0 < r < a_-$ and (ii) a thin surface layer $a_- < r < a_+$. In the limit of infinitesimal surface thickness, a_- approaches a_+. The interior pressure P_0 is uniform and the interior current density is zero so both $\mathbf{J} \times \mathbf{B}$ and ∇P are zero in the interior. Thus, finite current density exists *only* in the surface layer and everywhere else the magnetic field is a vacuum field. The plasma is also assumed to be surrounded by a perfectly conducting wall located at a radius b where $b > a$. MHD equilibrium, i.e., Eq. (10.60), can be written as

$$0 = -\mu_0 \nabla P_0 - \nabla\left(\frac{B_0^2}{2}\right) + \mathbf{B}_0 \cdot \nabla \mathbf{B}_0. \tag{10.141}$$

Because $\nabla \cdot \mathbf{B}_0 = r^{-1}\partial/\partial r\,(rB_{0r}) = 0$ for an azimuthally symmetric, axially uniform field and because B_r cannot be singular, B_{0r} must vanish everywhere. Since all equilibrium quantities depend on r only, the third term in Eq. (10.141) can be expanded

$$\mathbf{B}_0 \cdot \nabla \mathbf{B}_0 = \left(B_{0\theta}(r)\hat{\theta} + B_{0z}(r)\hat{z}\right) \cdot \nabla\left(B_{0\theta}(r)\hat{\theta} + B_{0z}(r)\hat{z}\right) = B_{0\theta}^2 \hat{\theta} \cdot \nabla \hat{\theta} = -\frac{B_{0\theta}^2}{r}\hat{r}. \tag{10.142}$$

Thus, Eq. (10.141) only has r components and so reduces to

$$0 = -\mu_0 \frac{\partial P_0}{\partial r} - \frac{\partial}{\partial r}\left(\frac{B_0^2}{2}\right) - \frac{B_{0\theta}^2}{r}. \tag{10.143}$$

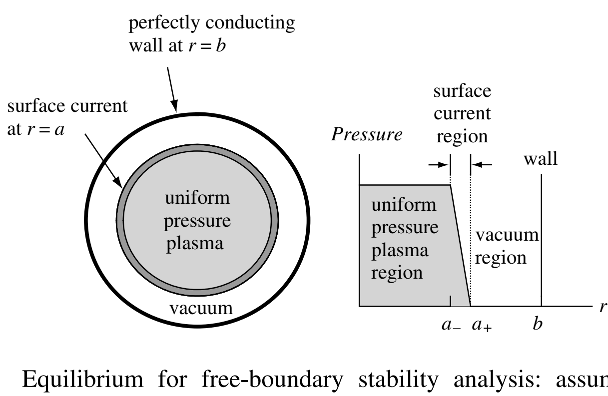

Fig. 10.8 Equilibrium for free-boundary stability analysis: assumed cross-section (left) and assumed pressure profile (right).

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On integrating across the surface layer, this gives the pressure balance relation

$$P_0 + \frac{B_{0pz}^2}{2\mu_0} = \frac{B_{0v\theta}^2 + B_{0vz}^2}{2\mu_0}, \tag{10.144}$$

where the subscript p refers to the inner side (i.e., plasma side) of the surface layer and the subscript v refers to the outer side (i.e., vacuum side). The reason no $B_{0p\theta}^2$ term appears on the left-hand side of Eq. (10.144) is that there are no interior plasma currents (from Ampère's law $2\pi a B_{0p\theta} = \mu_0 I = 0$).

The location of the equilibrium surface can be described in a formal mathematical fashion by the function $r = a$ or, equivalently, by the surface-defining equation

$$S_0(r) = r - a = 0. \tag{10.145}$$

Using this formalism, an incompressible perturbation can be characterized as a harmonic deformation of the plasma surface such that $r = a + \xi_r \mathrm{e}^{\mathrm{i}m\theta + \mathrm{i}kz}$. The equation describing the perturbed surface is thus

$$S(r, \theta, z) = r - a - \xi_r \mathrm{e}^{\mathrm{i}m\theta + \mathrm{i}kz} = 0, \tag{10.146}$$

where, without loss of generality, $\xi_r > 0$ is assumed. The equilibrium magnetic field has $B_{0r} = 0$ so no equilibrium magnetic field lines penetrate the surface. This property can be stated equivalently as the equilibrium magnetic field being tangential to the surface or, in a more abstract mathematical way, as $\mathbf{B}_0 \cdot \nabla S_0 = 0$. Since the magnetic field is assumed frozen into the plasma, the magnetic field must *continue* to be tangential to the surface even when the surface becomes deformed from its equilibrium shape. Thus, the condition

$$\mathbf{B} \cdot \nabla S = 0 \tag{10.147}$$

must be satisfied at all times where ∇S is in the direction normal to the surface; this is essentially a statement that no magnetic field line penetrates the surface even when the surface becomes deformed by the instability.

A quantity f at the perturbed surface (denoted by the subscript ps) can be expressed as

$$f_{ps} = f_0 + f_1 + \boldsymbol{\xi} \cdot \nabla f_0, \tag{10.148}$$

where the the middle term is the absolute first-order change and the last term is the convective term due to the motion of the surface.

In order for the configuration to be stable, any perturbation must generate a restoring force, which pushes the perturbed surface back to its equilibrium location. The competition between destabilizing and restoring forces is found by integrating Eq. (10.141) across the perturbed surface. This integration gives Eq. (10.144) evaluated at the perturbed surface, i.e., using Eq. (10.148) to take

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into account the effect of the perturbation of the surface. Since ξ_r was assumed positive (outward bulge at $\theta = 0$, $z = 0$) the perturbed configuration will be stable if the effective pressure on the perturbed surface exterior exceeds the effective pressure on the perturbed surface interior, i.e., if

$$\left[\frac{B_{v\theta}^2 + B_{vz}^2}{2\mu_0}\right]_{\text{perturbed sfc}} > \left[P + \frac{B_{pz}^2}{2\mu_0}\right]_{\text{perturbed sfc}} \Longrightarrow \text{stable.} \tag{10.149}$$

In this case, the restoring force pushes back the bulge and makes the system revert to its equilibrium condition. Subtracting the equilibrium pressure balance relation Eq. (10.144) from Eq. (10.149) gives

$$\frac{B_{0v\theta}B_{1v\theta} + B_{0vz}B_{1vz}}{\mu_0} + \xi_r \frac{\partial}{\partial r}\left[\frac{B_{0v\theta}^2 + B_{0vz}^2}{2\mu_0}\right] > P_1 + \frac{B_{0pz}B_{1pz}}{\mu_0} + \xi_r \frac{\partial}{\partial r}\left[P_0 + \frac{B_{0pz}^2}{2\mu_0}\right], \tag{10.150}$$

where all quantities are evaluated at $r = a$.

Equation (10.150) can be simplified considerably because of the following relationships:

1. There are no currents in either the plasma interior or the external vacuum so
$$\frac{\partial B_{0pz}}{\partial r} = \frac{\partial B_{0vz}}{\partial r} = 0. \tag{10.151}$$
2. The adiabatic pressure equation gives
$$P_1 + \boldsymbol{\xi} \cdot \nabla P_0 + \gamma P_0 \nabla \cdot \boldsymbol{\xi} = 0. \tag{10.152}$$
Since $\nabla \cdot \boldsymbol{\xi} = 0$ has been assumed (i.e., incompressibility), the adiabatic relation reduces to
$$P_1 + \xi_r \partial P_0 / \partial r = 0. \tag{10.153}$$
3. From Ampère's law it is seen that $B_{0v\theta}^2 \sim r^{-2}$ so $\left[\partial B_{0v\theta}^2 / \partial r\right]_{r=a} = -2B_{0v\theta}^2/a$; thus Eq. (10.150) simplifies to
$$B_{0v\theta}B_{1v\theta} + B_{0vz}B_{1vz} - \frac{\xi_r}{a} B_{0v\theta}^2 > B_{0pz}B_{1pz} \Longrightarrow \text{stable,} \tag{10.154}$$
where again all fields are evaluated at $r = a$.

To simplify the algebra all fields are now normalized to $B_{0v\theta}(a)$. The normalized fields are denoted by a bar on top and are

$$\bar{B}_{0vz} = \frac{B_{0vz}}{B_{0v\theta}(a)}, \quad \bar{B}_{0pz} = \frac{B_{0pz}}{B_{0v\theta}(a)},$$
$$\bar{B}_{1vz} = \frac{B_{1vz}}{B_{0v\theta}(a)}, \quad \bar{B}_{1v\theta} = \frac{B_{1v\theta}}{B_{0v\theta}(a)}, \quad \bar{B}_{1pz} = \frac{B_{1pz}}{B_{0v\theta}(a)} \tag{10.155}$$

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so, when expressed as a relationship between normalized fields, Eq. (10.154) becomes

$$\bar{B}_{1v\theta}+\bar{B}_{0vz}\bar{B}_{1vz}-\frac{\xi_r}{a}>\bar{B}_{0pz}\bar{B}_{1pz}\Longrightarrow \text{stable.} \tag{10.156}$$

What remains to be done is express all first-order magnetic fields in terms of ξ_r. This is relatively easy because the current is confined to the surface layer so the magnetic field is a vacuum magnetic field everywhere except exactly on the surface.

Since the magnetic field is a vacuum field for both $0<r<a$ and $a<r<b$, the normalized fields under consideration here must be of the form $\bar{\mathbf{B}}=\nabla\chi$, where $\nabla^2\chi=0$. This is true in both the plasma and vacuum regions but, due to the surface current at the vacuum–plasma interface, a jump discontinuity between the vacuum and plasma tangential magnetic fields occurs at $r=a$. Because the perturbed fields were assumed to have an $\exp(\mathrm{i}m\theta+\mathrm{i}kz)$ dependence, the Laplacian becomes

$$\nabla^2\chi=\frac{\mathrm{d}^2\chi(r)}{\mathrm{d}r^2}+\frac{1}{r}\frac{\mathrm{d}\chi(r)}{\mathrm{d}r}-\left(\frac{m^2}{r^2}+k^2\right)\chi(r)=0, \tag{10.157}$$

which has solutions consisting of the modified Bessel functions, $I_{|m|}(|k|r)$ and $K_{|m|}(|k|r)$. Absolute value signs have been used here to avoid possible confusion for situations to be encountered later where m or k could be negative. The $I_{|m|}(|k|r)$ function is finite at $r=0$ but diverges at $r=\infty$, while the opposite is true for $K_{|m|}(|k|r)$.

The surface current causes a discontinuity in the tangential components of the perturbed magnetic field. Hence, different solutions must be used in the respective plasma and vacuum regions and then these solutions must be related to each other in such a way as to satisfy the requirements of this discontinuity. The forms of the plasma and vacuum solutions are also constrained by the respective boundary conditions at $r=0$ and at $r=b$. In particular, the $K_{|m|}(|k|r)$ solution is not allowed in the plasma because of the constraint that the magnetic field must be finite at $r=0$; inside the plasma the solution must therefore be of the form

$$\chi=\alpha I_{|m|}(|k|r)\quad \text{in plasma region } 0\leq r\leq a_-. \tag{10.158}$$

On the other hand, both the $I_{|m|}(|k|r)$ and $K_{|m|}(|k|r)$ solutions are permissible in the vacuum region and so the vacuum region solution is of the general form

$$\chi=\beta_1 I_{|m|}(|k|r)+\beta_2 K_{|m|}(|k|r)\quad \text{in vacuum region } a_+\leq r\leq b. \tag{10.159}$$

The objective now is to express the coefficients α, β_1, and β_2 in terms of ξ_r so all components of the perturbed magnetic field can be expressed as a function of ξ_r, both in the plasma and in the vacuum.

The functional dependence of these coefficients is determined by considering boundary conditions at the wall and then at the plasma–vacuum interface:

1. Wall: since the wall is perfectly conducting it is a flux conserver. There is no radial magnetic field initially and so there is no flux linking each small patch of the wall. Thus $\bar{B}_{1vr}(b)$ must vanish at the wall in order to maintain zero flux at each patch of the wall. Using Eq. (10.159) to calculate the perturbed radial magnetic field $\bar{B}_{1vr} = \partial\chi/\partial r$ at the wall, setting $\bar{B}_{1vr}(b) = 0$ implies
$$\beta_1 I'_{|m|}(|k|b) + \beta_2 K'_{|m|}(|k|b) = 0, \tag{10.160}$$
which may be solved for β_2 to obtain
$$\beta_2 = -\beta_1 \frac{\hat{I}'_{|m|}}{\hat{K}'_{|m|}}; \tag{10.161}$$
here the circumflex means the modified Bessel function is evaluated at the wall, i.e., with its argument set to $|k|b$. Equation (10.159) can then be recast as
$$\chi = \beta\left[I_{|m|}(|k|r)\hat{K}'_{|m|} - \hat{I}'_{|m|}K_{|m|}(|k|r)\right], \tag{10.162}$$
where $\beta = \beta_1/\hat{K}'_{|m|}$. The wall boundary condition has reduced the number of independent coefficients by one.
2. Vacuum side of plasma–vacuum interface: here Eq. (10.147) is linearized to obtain
$$\mathbf{B}_1 \cdot \nabla S_0 + \mathbf{B}_0 \cdot \nabla S_1 = 0 \tag{10.163}$$
or using Eqs. (10.145) and (10.146)
$$\bar{B}_{1vr} - \mathrm{i}\left(\frac{m}{a} + k\bar{B}_{0vz}\right)\xi_r = 0. \tag{10.164}$$
On the vacuum side Eq. (10.162) gives
$$\bar{B}_{1vr} = |k|\beta\left[I'_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K'_{|m|}\right], \tag{10.165}$$
where omission of the argument means the modified Bessel function is evaluated at $r = a$. Substitution of Eq. (10.165) into Eq. (10.164) gives
$$\beta = \frac{\mathrm{i}\left(\dfrac{m}{a} + k\bar{B}_{0z}\right)}{|k|\left[I'_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K'_{|m|}\right]}\xi_r \tag{10.166}$$
so the complete vacuum field can now be expressed in terms of ξ_r.
3. Plasma side of plasma–vacuum interface: the plasma-side version of Eq. (10.163) gives
$$\bar{B}_{1pr} - \mathrm{i}k\bar{B}_{0pz}\xi_r = 0 \tag{10.167}$$

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since $B_{0p\theta}$ vanishes inside the plasma. From Eq. (10.158) the perturbed radial magnetic field on the plasma side of the interface is

$$\bar{B}_{1pr} = |k|\alpha I'_{|m|}. \tag{10.168}$$

The plasma-side version of Eq. (10.164) is thus

$$\alpha = \frac{\mathrm{i}k\bar{B}_{0pz}}{|k|I'_{|m|}}\xi_r \tag{10.169}$$

and so the plasma fields can now also be expressed in terms of ξ_r.

The stability condition, Eq. (10.156), can be written in terms of α and β to obtain

$$\beta\left(\frac{\mathrm{i}m}{a} + \mathrm{i}k\bar{B}_{0vz}\right)\left(I_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K_{|m|}\right) - \frac{\xi_r}{a} > \alpha\bar{B}_{0pz}\mathrm{i}kI_{|m|}. \tag{10.170}$$

Substituting for α and β and rearranging the order gives

$$|k|a\bar{B}^2_{0pz}\left[\frac{I_{|m|}}{I'_{|m|}}\right] - \frac{(m+ka\bar{B}_{0vz})^2}{|k|a}\left[\frac{I_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K_{|m|}}{I'_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K'_{|m|}}\right] > 1 \Longrightarrow \text{stable.} \tag{10.171}$$

If we introduce the normalized pressure

$$\bar{P}_0 = \frac{2\mu_0 P_0}{B^2_{0v\theta}(a)}, \tag{10.172}$$

the MHD equilibrium, Eq. (10.144), can be written in normalized form as

$$\bar{P}_0 + \bar{B}^2_{0pz} = 1 + \bar{B}^2_{0vz}. \tag{10.173}$$

Substitution for $\bar{B}^2_{0pz}$ into Eq. (10.171) and rearranging the order of the second term gives

$$|k|a\left[1+\bar{B}^2_{0vz} - \bar{P}_0\right]\left[\frac{I_{|m|}}{I'_{|m|}}\right] + \frac{(m+ka\bar{B}_{0vz})^2}{|k|a}\left[\frac{-I_{|m|}\hat{K}'_{|m|} + \hat{I}'_{|m|}K_{|m|}}{I'_{|m|}\hat{K}'_{|m|} - \hat{I}'_{|m|}K'_{|m|}}\right] > 1, \tag{10.174}$$

which is the general requirement for stability of a configuration having a specified internal pressure, vacuum axial field, plasma radius, and wall radius.

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Conditions for sausage and kink instability

Let us now examine the effect of the various factors in Eq. (10.174). For large argument, the modified Bessel functions have the asymptotic form

$$\lim_{s\to\infty} \quad I_{|m|}(s) \to \frac{1}{\sqrt{2\pi s}}\mathrm{e}^{s}; \quad \lim_{s\to\infty} \quad K_{|m|}(s) \to \sqrt{\frac{\pi}{2s}}\mathrm{e}^{-s} \tag{10.175}$$

so if the wall radius goes to infinity, the factor

$$\left[\frac{-I_{|m|}\hat{K}'_{|m|}+\hat{I}'_{|m|}K_{|m|}}{I'_{|m|}\hat{K}'_{|m|}-\hat{I}'_{|m|}K'_{|m|}}\right] \to -\frac{K_{|m|}}{K'_{|m|}} = \text{positive-definite.} \tag{10.176}$$

Since bringing the wall closer is stabilizing, the factor

$$\frac{-I_{|m|}\hat{K}'_{|m|}+\hat{I}'_{|m|}K_{|m|}}{I'_{|m|}\hat{K}'_{|m|}-\hat{I}'_{|m|}K'_{|m|}}$$

will always be positive-definite. In particular, if $b \to a$ then $I'_{|m|}\hat{K}'_{|m|} \to \hat{I}'_{|m|}K'_{|m|}$, in which case the wall stabilization becomes arbitrarily large.

As $|k|a \to \infty$ the left-hand side of Eq. (10.174) becomes infinite. Thus, the configuration is stable with respect to modes having short axial wavelengths. This is because short wavelength perturbations cause more bending of the magnetic field than a long wavelength perturbation and so require more energy.

We therefore focus attention on modes with a *long* axial wavelength (i.e., modes with a small k) since these offer the only possibility of instability. The analysis can be subdivided into specific cases, such as $m = 0$, $|m| = 1$, $|m| > 1$, close-fitting wall, no wall, low pressure, high pressure, etc. Before getting into the details, let us take a broader look at the effect of the various terms in Eq. (10.174). Since $I_{|m|}/I'_{|m|} > 0$, we see that increasing $\bar{P}_0$ is destabilizing, whereas increasing $\bar{B}^2_{0vz}$ is stabilizing. Also, if $m + ka\bar{B}_{0vz} = 0$ the second term vanishes, leading to *reduced* stability; by defining the wavevector as $\mathbf{k} = (m/a)\hat{\theta} + k\hat{z}$ it is seen that $m + ka\bar{B}_{0vz} = 0$ corresponds to having $\mathbf{k}\cdot\mathbf{B}_0 = 0$ on the surface.

Sausage modes

The $m = 0$ modes are the sausage instabilities and here Eq. (10.174) reduces to

$$\left[1+\bar{B}^2_{0vz}-\bar{P}_0\right]\left[\frac{I_0}{I'_0}\right] + \bar{B}^2_{0vz}\left[\frac{-I_0\hat{K}'_0+\hat{I}'_0K_0}{I'_0\hat{K}'_0-\hat{I}'_0K'_0}\right] > \frac{1}{|k|a} \Longrightarrow \text{stable.} \tag{10.177}$$

For a given normalized plasma pressure and normalized wall radius b/a, this expression can be used to make a stability plot of $\bar{B}^2_{0vz}$ versus $|k|a$. Since the wall always provides stabilization if brought in close enough, let us consider

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situations where there is no wall (i.e., $b \to \infty$), in which case the stability condition reduces to

$$\left[1+\bar{B}_{0vz}^2-\bar{P}_0\right]\left[\frac{I_0}{I_0'}\right]+\bar{B}_{0vz}^2\left[\frac{-K_0}{K_0'}\right] > \frac{1}{|k|a} \Longrightarrow \text{stable.} \tag{10.178}$$

For small arguments, the modified Bessel functions of order zero have the asymptotic values

$$I_0(s) \simeq 1+\frac{s^2}{4}; \quad K_0(s) \simeq -\ln s \tag{10.179}$$

so the stability criterion becomes

$$\bar{B}_{0vz}^2\left[1-k^2a^2\ln\left(|k|a\right)\right] > \bar{P}_0 \Longrightarrow \text{stable.} \tag{10.180}$$

This gives a simple criterion for how much $\bar{B}_{0vz}^2$ is required to stabilize a given plasma pressure against sausage instabilities. The logarithmic term is stabilizing for $|k|a<1$ but is destabilizing for $|k|a>1$; however, this region of instability is limited because we showed that very large $|k|a$ is stable.

Kink modes

The finite m modes are the kink modes. It was shown earlier that large $|k|a$ is stable so again we confine attention to small $|k|a$. In addition, we again let the wall location go to infinity to simplify the analysis. The stability condition now reduces to

$$|k|a\left[1+\bar{B}_{0vz}^2-\bar{P}_0\right]\left[\frac{I_{|m|}}{I_{|m|}'}\right]+\frac{(m+ka\bar{B}_{0vz})^2}{|k|a}\left[\frac{-K_{|m|}}{K_{|m|}'}\right] > 1 \Longrightarrow \text{stable.} \tag{10.181}$$

For $m \neq 0$ the small argument asymptotic limits of the modified Bessel functions are

$$I_{|m|}(s) \simeq \frac{1}{|m|!}\left(\frac{s}{2}\right)^{|m|}; \quad K_{|m|}(s) \simeq \frac{|m-1|!}{2}\left(\frac{s}{2}\right)^{-|m|} \tag{10.182}$$

so the stability condition becomes

$$k^2a^2\left[1+\bar{B}_{0vz}^2-\bar{P}_0\right]+(m+ka\bar{B}_{0vz})^2 > |m| \Longrightarrow \text{stable,} \tag{10.183}$$

which is a quadratic equation in ka. Let us consider plasmas where $\bar{B}_{0vz}^2 \gg 1$ and $\bar{B}_{0vz}^2 \gg \bar{P}_0$; this corresponds to a low-beta plasma where the externally imposed axial field is much stronger than the field generated by the internal plasma currents (tokamaks are in this category). Let us define

$$x = ka\bar{B}_{0vz} \tag{10.184}$$

so the stability condition becomes

$$x^2+(m+x)^2 > |m| \Longrightarrow \text{stable.} \tag{10.185}$$

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Without loss of generality x can be assumed to be positive, in which case instability occurs only when m is negative. Thus, Eq. (10.185) can be written as

$$2x^2 - 2|m|x + m^2 - |m| > 0 \Longrightarrow \text{stable.} \tag{10.186}$$

The threshold for instability occurs when the left-hand side of Eq. (10.186) vanishes, i.e., at the roots of the left-hand side. These roots are

$$x = \frac{|m| \pm \sqrt{2|m| - m^2}}{2}. \tag{10.187}$$

Since the left-hand side of Eq. (10.186) goes to positive infinity for $|x| \to \infty$, the left-hand side is negative only in the region between the two roots. Thus, the plasma is unstable only if $|x|$ lies between the two roots. The stability condition is that $|x|$ must lie outside the region between the two roots, i.e., for stability we must have

$$\begin{aligned} x &> \frac{|m| + \sqrt{2|m| - m^2}}{2} \quad \text{or} \\ x &< \frac{|m| - \sqrt{2|m| - m^2}}{2} \quad . \end{aligned} \tag{10.188}$$

For $m = -1$ modes this gives the stability condition

$$x > 1. \tag{10.189}$$

In un-normalized quantities and using $k = 2\pi/L$, where L is the axial length of the system, the stability condition is

$$\frac{2\pi a}{L} \frac{B_{0z}}{B_{0\theta}} > 1; \tag{10.190}$$

this is known as the Kruskal–Shafranov kink stability criterion (Kruskal *et al.* 1958, Shafranov 1958).

For $m = -2$ modes, the two roots coalesce at $x = 1$ and so Eq. (10.189) also gives stability. For $m \geq 3$, the argument of the square root in Eq. (10.187) is negative so there is no region of instability.

In toroidal devices such as tokamaks, the axial wavenumber corresponds to the toroidal wavenumber since the dominant magnetic field is in the toroidal direction, i.e., $B_{0z} \to B_{0\phi}$, where ϕ is the toroidal angle. The axial wavenumber k becomes n/R, where n is the toroidal mode number. Since long axial wavelengths are the most unstable we assume $n = 1$ to have the worst-case scenario. The Kruskal–Shafranov kink stability condition then becomes

$$q = \frac{aB_{0\phi}}{RB_{0\theta}} > 1, \tag{10.191}$$

where q is called the safety factor. In order to avoid kink instability, tokamaks typically operate with $q \sim 3-4$ at the wall and q slightly above unity at the magnetic axis; this q condition is one of the most important design criteria for tokamaks since it dictates the size of the large and expensive toroidal field system.

10.10 Assignments

1. Interchange instabilities and volume per unit flux. Another way to consider pressure-driven instabilities is to calculate the consequence of interchanging two flux tubes having the same magnetic flux. The plasma moves across the magnetic field in such a way that the frozen-in flux condition is maintained. This interchange will not change the magnetic energy since the magnetic field is unaffected. However, if the flux tubes contain finite-pressure plasma and the volumes of the two flux tubes differ, then the interchange will result in compression of the plasma in the flux tube that initially had the larger volume and expansion of plasma in the flux tube that initially had the smaller volume. The former requires work on the plasma and the latter involves work by the plasma. If net work must be done on the plasma to effect the interchange, then the interchange is stable and vice versa. In a magnetic confinement configuration, the interior region has higher pressure than the exterior region. Thus, the question is whether interchanging a high pressure, interior region flux tube with a low pressure, exterior region flux tube requires positive or negative work.
   (a) Show that the volume per unit flux in a flux tube is given by
   $$V' = \oint \frac{\mathrm{d}l}{B},$$
   where the contour is over the length of the flux tube. Hint: use $BA = const.$ on a flux tube.
   (b) Show that instability corresponds to V' increasing on going from interior to exterior.
   (c) Consider the magnetic field external to a current-carrying straight wire. How does V' scale with distance from the wire? Would a plasma confined by such a magnetic field be stable or unstable to interchanges? Is this result consistent with the concepts of good and bad curvature?
2. Work through the algebra of the magnetic energy principle and verify Eqs. (10.113), (10.114), (10.115), and (10.116).
3. Show that the force on a plasma in an *arched magnetic field* with fixed ends tends to push the plasma towards the shape of a vacuum magnetic field arch having the same boundary conditions. Under what circumstances will the force (i) increase the major radius of an arched magnetic field, (ii) decrease the major radius, (iii) leave the major radius unchanged? Show that, in contrast, the force on a plasma containing an *arched current* always tends to increase the major radius of the arched current.

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4. Show that if the wall radius $b \to \infty$, then Eq. (10.174) reduces to the condition

$$F(x, \bar{B}_{0vz}, \bar{P}_0) = x\bar{B}_{pvz}^2 \left[\frac{I_m(x)}{I'_m(x)} \right] - \frac{(m + x\bar{B}_{0vz})^2}{x} \frac{K_m(x)}{K'_m(x)} - 1 > 0 \text{ for stability,}$$

   where $x = ka$. By assuming $\bar{B}_{pvz} = \bar{B}_{0vz}$ and evaluating the modified Bessel functions numerically, plot this expression in a parameter space where the vertical axis is x and the horizontal axis is $B_{0v\theta}/B_{0vz} = 1/\bar{B}_{0vz}$. In particular, shade the regions where $F < 0$ to indicate the regions of instability. Do this for both $m = 0$ and for $m = -1$ to show the regions of parameter space where kinks and sausages are unstable. For a given k show how raising the axial current would lead to kink or sausage instability. Which instability happens first (kink or sausage)?
5. Are kink instabilities diamagnetic or paramagnetic with respect to the axial field? To find this, consider the kinked current as a solenoid and determine whether the orientation of the solenoid is such as to increase or decrease the initial B_z field.
6. According to Eq. (10.139), current driven instabilities cause the plasma to relax to a situation where $\mu_0 \mathbf{J} = \lambda \mathbf{B}$.
   (a) Show that a class of solutions to $\mu_0 \mathbf{J} = \lambda \mathbf{B}$ can be found if

$$\mathbf{B} = \lambda \nabla \psi \times \hat{z} + \nabla \times (\nabla \psi \times \hat{z}),$$

   where $\nabla^2 \psi + \lambda^2 \psi = 0$.
   (b) Suppose $\psi(x, z) = f(z)\cos(kx)$. What is the form of $f(z)$? Show that solutions that decay in z are only possible for a certain range of λ^2.
   (c) Consider a two-dimensional force-free solar coronal loop. Let the $z = 0$ plane denote the solar surface. Calculate the functional form of B_x and B_z using the solution in part (a) above.
   (d) Sketch the shape of the field line going from $x = -\pi/k$ to $x = +\pi/k$ in the $z = 0$ plane. Do this for a sequence of increasing values of λ^2. What happens to the shape of the field line as λ^2 is increased? How is current related to λ?

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